Electronic Transport in Mesoscopic Systems

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Contents

Chapter 1

Preliminary concepts

1.1 Two-dimesional electron gas

The drift velocity of electrons under an applied electric field $\boldsymbol{E}$ is given by

$$\left[\frac{\mathrm{d}p}{\mathrm{d}t}\right]_{scattering} = \left[\frac{\mathrm{d}p}{\mathrm{d}t}\right]_{field} \tag{1.1.1}$$

Hence, (τ_m : $momentumrelaxationtime$)

$$\frac{m\boldsymbol{v}_d}{\tau_m} = e\boldsymbol{E} \Rightarrow \boldsymbol{v}_d = \frac{e\tau_m}{m}\boldsymbol{E} \tag{1.1.2}$$

The mobility μ is defined as the ratio of the drift velocity to the applied electric field, i.e.,

$$\mu = \frac{v_d}{E} = \frac{e\tau_m}{m} \tag{1.1.3}$$

1.2 Effective mass, density of states etc.

The dynamics of electron in the conduction band can be described by an equation of the form

$$\left[E_c + \frac{(\mathrm{i}\hbar\nabla + e\boldsymbol{A})^2}{2m} + U(\boldsymbol{r})\right]\psi(\boldsymbol{r}) = E\psi(\boldsymbol{r}) \tag{1.2.1}$$

$U(\boldsymbol{r})$ is the potential energy due to space-charge etc., and $\boldsymbol{A}$ is the vector potential associated with magnetic field $\boldsymbol{B} = \nabla \times \boldsymbol{A}$. Considering a homogeneous material with $U(\boldsymbol{r}) = 0$, and $\boldsymbol{A} = 0$, and $E_c =$ constant, the wavefunction solutions are plane waves of the form

$$\psi(\boldsymbol{r}) = \frac{1}{\sqrt{V}}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} \tag{1.2.2}$$

and not that of Bloch waves

$$\psi(\boldsymbol{r}) = u_{\boldsymbol{k}}(\boldsymbol{r})\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}} \tag{1.2.3}$$

1.2.1 Subbands

Consider the 2-DEG. The electrons are free to propagate in the x-y plane but are confined by some potential $U(z)$ in the z-direction. The wavefunction in such a case(with vector potential $\boldsymbol{A} = 0$) can be written as

$$\psi(\boldsymbol{r}) = \phi_n(z)\mathrm{e}^{\mathrm{i}(k_x x + k_y y)} \tag{1.2.4}$$

with the dispersion relation

$$E = E_c + \epsilon_n + \frac{\hbar^2\left(k_x^2 + k_y^2\right)}{2m} \tag{1.2.5}$$

The index n numbers the different subbands each having wavefunction $\phi_n(z)$ and a cut-off energy ϵ_n. Usually at low tempertures with low carrier densities, only the lowest subband $n = 1$ is occupied and the higher subbands are empty. We can then ignore the z-direction altogether and treat the system as a purely 2D system in x-y plane. We will use the following equation:

$$\left[E_s + \frac{(\mathrm{i}\hbar\nabla + e\boldsymbol{A})^2}{2m}\right]\psi(x,y) = E\psi(x,y) \tag{1.2.6}$$

where $E_s = E_c + \epsilon_1$ is the conduction band edge of the lowest subband.

1.2.2 Band diagrams

Setting $U = 0$ and $\boldsymbol{A} = 0$, The eigenfunctions normalized to an area S have the form

$$\psi(x,y) = \frac{1}{\sqrt{S}}\exp(\mathrm{i}k_x x)\exp(\mathrm{i}k_y y) \tag{1.2.7}$$

with the dispersion relation

$$E = E_s + \frac{\hbar^2\left(k_x^2 + k_y^2\right)}{2m} \tag{1.2.8}$$

1.2.3 Density of states

We denote the total number of states having an energy less than E as $N_T(E)$. We count the number of states contained within a circle of radius k, where

$$E = E_s + \frac{\hbar^2 k^2}{2m} \Rightarrow k = \sqrt{\frac{2m(E - E_s)}{\hbar^2}} \tag{1.2.9}$$

using peroidic boundary conditions, the allowed values of k_x and k_y are given by

$$k_x = \frac{2\pi}{L_x}n_x, \quad k_y = \frac{2\pi}{L_y}n_y \tag{1.2.10}$$

where n_x and n_y are integers. Each allowed state occupies an area of

$$\Delta k_x \Delta k_y = \frac{(2\pi)^2}{L_x L_y} = \frac{(2\pi)^2}{S} \tag{1.2.11}$$

while the area of the circle is πk^2. Thus the total number of states having energy less than E is given by

$$N_T(E) = g_s\frac{\pi k^2}{(2\pi)^2/S} = g_s\frac{S}{4\pi}k^2 = \frac{m}{\pi\hbar^2}S(E - E_s) \tag{1.2.12}$$

where $g_s = 2$ is the spin degeneracy factor. The density of states(DOS) per unit area is given by

$$N(E) = \frac{1}{S}\frac{\mathrm{d}N_T(E)}{\mathrm{d}E} = \frac{m}{\pi\hbar^2}\theta(E - E_s) \tag{1.2.13}$$

1.2.4 Degenerate and non-degenerate conductors

At equilibrium, the available states in a conductor are filled up according to the Fermi function

$$f_0(E) = \frac{1}{1 + \exp(E - E_f)/k_B T} \tag{1.2.14}$$

In high temperture or non-degenerate limit($\exp[E_s - E_f]/k_BT \gg 1$) the Fermi function can be approximated as

$$f_0(E) \approx \exp(-(E - E_f)/k_BT) \tag{1.2.15}$$

In low temperture or degenerate limit($\exp[E_s - E_f]/k_BT \ll 1$), the Fermi function can be approximated as

$$f_0(E) \approx \theta(E_f - E) \tag{1.2.16}$$

The equilibrium electron density n_s(per unit area) is given by

$$n_s = \int N(E) f_0(E) \mathrm{d}E \tag{1.2.17}$$

For degenerate conductors it is easy to perform the integration:

$$\begin{aligned} n_s &= \int \theta(E_f - E)\theta(E - E_s)\frac{m}{\pi\hbar}\mathrm{d}E \\ &= \frac{m}{\pi\hbar^2}(E_f - E_s) = N_s(E_f - E_s), \quad N_s = \frac{m}{\pi\hbar^2} \end{aligned} \tag{1.2.18}$$

The Fermi wavenumber k_f is given by

$$E_f - E_s = \frac{\hbar^2 k_f^2}{2m} \Rightarrow k_f = \sqrt{2\pi n_s} = \sqrt{\frac{2m(E_f - E_s)}{\hbar^2}} \tag{1.2.19}$$

The corresponding Fermi velocity v_f is $v_f = \hbar k_f/m$.

1.3 Characteristic lengths

1.3.1 wavelength

The Fermi wavelength is

$$\lambda_f = \frac{2\pi}{k_f} = \sqrt{\frac{2\pi}{n_s}} \tag{1.3.1}$$

1.3.2 Mean free path

The momentum relaxation time τ_m is relateed to the collisition time τ_c by a relation of the form

$$\frac{1}{\tau_m} \to \frac{1}{\tau_c}\alpha_m \tag{1.3.2}$$

where the factor α_m (lying between 0 and 1) denotes the "effectiveness" of an individual collision in destroying momentum. The factor α_m is then very small so that the momentum relaxation time is much longer than the collision time. The mean free path l is given by

$$l = v_f \tau_m \tag{1.3.3}$$

where τ_m is the momentum relaxation time and v_f is the Fermi velocity.

1.3.3 Phase coherence length

In analogy with the momentum relaxation time we could write

$$\frac{1}{\tau_\phi} \to \frac{1}{\tau_c}\alpha_\phi \tag{1.3.4}$$

where α_ϕ is the "effectiveness" of an individual collision in destroying phase coherence. The phase coherence length L_ϕ is then given by

$$L_\phi = v_f \tau_\phi \tag{1.3.5}$$

The momentum relaxation time τ_m is usually much shorter than the phase relaxation time τ_ϕ since many collisions are required to randomize the phase of the electron wavefunction. After an interval of time τ_m the velocity is completely randomized so that the electronic trajectory over a time τ_ϕ can be viewed as a random walk with a step length of $v_f \tau_m$ and a total number of steps of τ_ϕ/τ_m. Since individual trajectory are in random directions(θ), the root mean square distance traveled by the electron in a particular direction is obtained by summing the squares of their lengths:

$$L_\phi^2 = \frac{\tau_\phi}{\tau_m}(v_f \tau_m)^2 \left\langle \cos^2\theta \right\rangle, \quad \left\langle \cos^2\theta \right\rangle = \int_0^{2\pi} \cos^2\theta \frac{\mathrm{d}\theta}{2\pi} = \frac{1}{2} \tag{1.3.6}$$

Hence,

$$L_\phi^2 = \frac{1}{2}(v_f \tau_m)^2 \frac{\tau_\phi}{\tau_m} = v_f^2 \tau_m \tau_\phi / 2 \tag{1.3.7}$$

The diffusion coefficient D is defined as

$$D = \frac{1}{2} v_f^2 \tau_m \tag{1.3.8}$$

Thus we have

$$L_\phi^2 = D\tau_\phi \tag{1.3.9}$$

1.4 Low-field magnetoresistance

1.4.1 Drude model

The drift velocity in the presence of both electric and magnetic fields is given by

$$\frac{m\boldsymbol{v}_d}{\tau_m} = e\left[\boldsymbol{E} + \boldsymbol{v}_d \times \boldsymbol{B}\right] \tag{1.4.1}$$

We can rewrite (1.4.1) in the form

$$\begin{bmatrix} m/e\tau_m & -B \\ B & m/e\tau_m \end{bmatrix} \begin{bmatrix} v_{dx} \\ v_{dy} \end{bmatrix} = \begin{bmatrix} E_x \\ E_y \end{bmatrix} \tag{1.4.2}$$

The current density $\boldsymbol{J} = -en_s\boldsymbol{v}_d$ is related to the electric field by the conductivity tensor σ as $\boldsymbol{J} = \sigma\boldsymbol{E}$. Thus we have

$$\begin{bmatrix} m/e\tau_m & -B \\ B & m/e\tau_m \end{bmatrix} \begin{bmatrix} J_x/en_s \\ J_y/en_s \end{bmatrix} = \begin{bmatrix} E_x \\ E_y \end{bmatrix} \tag{1.4.3}$$

Rearanging we obtain

$$\begin{bmatrix} E_x \\ E_y \end{bmatrix} = \sigma^{-1} \begin{bmatrix} 1 & -\mu B \\ \mu B & 1 \end{bmatrix} \begin{bmatrix} J_x \\ J_y \end{bmatrix} \tag{1.4.4}$$

where $\sigma = n_s e\mu$ is the conductivity at zero magnetic field and $\mu = e\tau_m/m$ is the mobility. The resistivity tensor $\rho = \sigma^{-1}$ is given by

$$\begin{bmatrix} E_x \\ E_y \end{bmatrix} \begin{bmatrix} \rho_{xx} & \rho_{xy} \\ \rho_{yx} & \rho_{yy} \end{bmatrix} \begin{bmatrix} J_x \\ J_y \end{bmatrix} \tag{1.4.5}$$

where

$$\rho_{xx} = \rho_{yy} = \frac{1}{\sigma}, \quad \rho_{xy} = -\rho_{yx} = \frac{\mu B}{\sigma} = B/n_s e \tag{1.4.6}$$

Thus this simple Drude model predicts that the longitudinal resistivity ρ_{xx} is independent of magnetic field while the Hall resistivity ρ_{xy} increases linearly with magnetic field.

1.5 High-field magnetoresistance

When the magnetic field is sufficiently strong such that the cyclotron frequency $\omega_c = eB/m$ becomes comparable to the collision frequency $1/\tau_m$, the simple Drude model breaks down. In this high-field regime, the electrons undergo cyclotron motion between collisions, leading to quantization of energy levels into Landau levels. The density of states becomes a series of delta functions at these Landau levels, significantly altering the transport properties of the system. This results in phenomena such as the quantum Hall effect, where the Hall resistivity exhibits plateaus at quantized values, and the longitudinal resistivity shows oscillatory behavior known as Shubnikov-de Haas oscillations.

1.5.1 Origin of SdH oscillations

The Sdh oscillations arise because at high magnetic fields, the step-like density of states associated with a 2-DEG

$$N(E) = \frac{m}{\pi\hbar^2}\theta(E - E_s) \tag{1.5.1}$$

breaks up into a series of delta functions spaced by the cyclotron energy $\hbar\omega_c$. where $\omega_c = eB/m$ is the cyclotron frequency.

$$N_s(E, B) \approx \frac{2eB}{h}\sum_{n=0}^{\infty}\delta(E - E_n), \quad E_n = E_s + (n + \frac{1}{2})\hbar\omega_c \tag{1.5.2}$$

As we change the magnetic field B, the energies of the landau levels change. For a given electron density n_s, the number of filled landau levels is given by

$$N = \frac{n_s h}{2eB} \tag{1.5.3}$$

The resistivity ρ_{xx} goes through maxima each time a landau level passes through the Fermi energy E_f. Hence the magnetic field values B_1 and B_2 corresponding to two successive maxima in ρ_{xx} are given by

$$\frac{n_s h}{2eB_1} - \frac{n_s h}{2eB_2} = 1 \tag{1.5.4}$$

so that

$$n_s = \frac{2e}{h}\frac{B_1 B_2}{B_2 - B_1} \tag{1.5.5}$$

Choose different magnetic field values corresponding to successive maxima in ρ_{xx} and use the above equation to calculate n_s.

1.5.2 Why do discrete states form at high magnetic fields?

Classically, an electron in a magnetic field moves in a circular orbit with cyclotron frequency $\omega_c = eB/m$, and the radius $r_c = v/\omega_c$. Quantum mechanically, the orbit must be an integer number (n) of de Broglie wavelengths:

$$2\pi r_c = nh/mv \tag{1.5.6}$$

We find the kinetic energy can only take discrete values given by $mv^2/2 = n\hbar\omega_c/2$. Hence we would expect that

$$E_n = E_s + \frac{n}{2}\hbar\omega_c \tag{1.5.7}$$

This is a little different from the correct answer from a proper quantum mechanical treatment, which gives

$$E_n = E_s + \left(n + \frac{1}{2}\right)\hbar\omega_c \tag{1.5.8}$$

These energy level E_n with different values are referred to as Landau levels. The density of states associated with each Landau level is given by

$$N_s(E,B) \approx N_0 \sum_{n=0}^{\infty} \delta\left(E - E_n\right), \quad N_0 = \frac{2eB}{h} \tag{1.5.9}$$

The correct prefactor N_0 can be obtained by noting that the magnetic field causes all the states in an energy range $\hbar\omega_c$ to be concentrated into a single Landau level. Since the density of states in the absence of magnetic field is $N_s = m/\pi\hbar^2$, we have

$$N_0 = N_s \hbar\omega_c = \frac{m}{\pi\hbar^2}\frac{\hbar eB}{m} = \frac{2eB}{h} \tag{1.5.10}$$

1.6 Transverse modes

Transverse mode or subbands are analogous to the transverse modes of electromagnetic waveguids. In narrow conductors. the different transverse modes are well seperated in energy and such conductors are often called electron waveguids. Consider a rectangular conductor that is uniform in the x-direction and has some transverse confining potential $U(y)$. The motion of electrons in such a conductor can be described by the equation

$$\left[E_c + \frac{(\mathrm{i}\hbar\nabla + e\boldsymbol{A})^2}{2m} + U(y)\right)]\psi(x,y) = E\psi(x,y) \tag{1.6.1}$$

We assume a constant magnetic field $\boldsymbol{B}$ along the z-direction. Choosing the Landau gauge $\boldsymbol{A} = (-By, 0, 0)$, we can write the wavefunction as

$$\left[E_s + \frac{(p_x + eBy)^2}{2m} + \frac{p_y^2}{2m} + U(y)\right]\psi(x,y) = E\psi(x,y) \tag{1.6.2}$$

The solutions can be written as

$$\psi(x,y) = \frac{1}{\sqrt{L}}\exp(\mathrm{i}kx)\chi(y) \tag{1.6.3}$$

where $\chi(y)$ satisfies the equation

$$\left[E_s + \frac{(\hbar k + eBy)^2}{2m} + \frac{p_y^2}{2m} + U(y)\right]\chi(y) = E\chi(y) \tag{1.6.4}$$

For a parabolic potential $U(y) = \frac{1}{2}m\omega_0^2 y^2$, the analytical solution is possible.

1.6.1 Confined electrons in zero magnetic field

The equation for $\chi(y)$ reduces to

$$\left[E_s + \frac{\hbar^2 k^2}{2m} + \frac{p_y^2}{2m} + U(y)\right]\chi(y) = E\chi(y) \tag{1.6.5}$$

The eigenfunctions and eigenfunctions are given by

$$\chi_{n,k}(y) = u_n(q), \quad q = y/y_0, \quad y_0 = \sqrt{\frac{\hbar}{m\omega_0}} \tag{1.6.6}$$

$$E(n,k) = E_s + \hbar\omega_0\left(n + \frac{1}{2}\right) + \frac{\hbar^2k^2}{2m}, \quad n = 0, 1, 2, ... \tag{1.6.7}$$

where $u_n(q) = \exp(-q^2/2)H_n(q)$, $H_n(q)$ being the nth Hermite polynomial. The velocity is obtained from the slope of the dispersion curve:

$$v_n(k) = \frac{1}{\hbar}\frac{\partial E(n,k)}{\partial k} = \frac{\hbar k}{m} \tag{1.6.8}$$

1.6.2 Unconfined electrons in non-zero magnetic field

The equation for $\chi(y)$ reduces to

$$\left[E_s + \frac{(\hbar k + eBy)^2}{2m} + \frac{p_y^2}{2m}\right]\chi(y) = E\chi(y) \tag{1.6.9}$$

which can be rewritten as

$$\left[E_s + \frac{p_y^2}{2m} + \frac{1}{2}m\omega_c^2(y+y_k)^2\right]\chi(y) = E\chi(y), \quad y_k = \frac{\hbar k}{eB}, \quad \omega_c = \frac{eB}{m} \tag{1.6.10}$$

The eigenfunctions and eigenvalues are given by

$$\chi_{n,k}(y) = u_n(q+q_k), \quad q = \sqrt{m\omega_c/\hbar}y, \quad q_k = \sqrt{m\omega_c/\hbar}y_k \tag{1.6.11}$$

The group velocity is vanished for eigenvalues are independent of k. The number of electrons which can fit into one Landau level equals to the 2D density of states times the cyclotron energy:

$$N = N_s S\hbar\omega_c = \frac{mS}{\pi\hbar^2}\frac{\hbar eB}{m} = \frac{eBS}{\pi\hbar} \tag{1.6.12}$$

Noting that the allowed values of k are given by $k = 2\pi n/L$, where n is an integer, the spacing between successive values of y_k is given by

$$\Delta y_k = \frac{\hbar\Delta k}{eB} = \frac{2\pi\hbar}{eBL} \tag{1.6.13}$$

Thus the total number of allowed values of y_k in a sample of width W is given by

$$N = g_s\frac{W}{\Delta y_k} = \frac{eBS}{\pi\hbar} \tag{1.6.14}$$

1.6.3 Confined electrons in non-zero magnetic field

When both the magnetic field and the confining parabolic potential are present, the equation for $\chi(y)$ becomes

$$\left[E_s + \frac{p_y^2}{2m} + \frac{(eBy+\hbar k)^2}{2m} + \frac{1}{2}m\omega_0^2 y^2\right]\chi(y) = E\chi(y) \tag{1.6.15}$$

This can be rewritten as

$$\left[E_s + \frac{p_y^2}{2m} + \frac{1}{2}m\frac{\omega_0^2\omega_c^2}{\omega_{c0}^2}y_k^2 + \frac{1}{2}m\omega_{c0}^2\left(y + \frac{\omega_c^2}{\omega_{c0}^2}y_k\right)^2\right]\chi(y) = E\chi(y) \tag{1.6.16}$$

where $\omega_{c0}^2 = \omega_c^2 + \omega_0^2$. The eigenfunctions and eigenvalues are given by

$$\chi_{n,k}(y) = u_n\left(q + \frac{\omega_c^2}{\omega_{c0}^2}q_k\right), \quad q = \sqrt{m\omega_{c0}/\hbar}y, \quad q_k = \sqrt{m\omega_{c0}/\hbar}y_k \tag{1.6.17}$$

$$E(n,k) = E_s + \hbar\omega_{c0}\left(n + \frac{1}{2}\right) + \frac{\hbar^2 k^2}{2m}\frac{\omega_0^2}{\omega_{c0}^2}, \quad n = 0,1,2,... \tag{1.6.18}$$

The group velocity is given by

$$v_n(k) = \frac{1}{\hbar}\frac{\partial E(n,k)}{\partial k} = \frac{\hbar k}{m}\frac{\omega_0^2}{\omega_{c0}^2} \tag{1.6.19}$$

The effect of the magnetic field is to increase the effective mass by

$$m^* = m\frac{\omega_{c0}^2}{\omega_0^2} = m\left(1 + \frac{\omega_c^2}{\omega_0^2}\right) \tag{1.6.20}$$

As the magnetic field increases, the dispersion relation look nealy flat, similar to the Landau levels. The spatial localtion of the eigenstates ψ_{nk} is centered around $y = -y_k$ where

$$y_k = \frac{\hbar k}{eB} \Rightarrow y_k = v_n(k)\frac{\omega_0^2 + \omega_c^2}{\omega_c\omega_0^2} \tag{1.6.21}$$

1.7 Drift velocity or Fermi velocity?

The current J in a homogeneous conductor is expressed as the product of the electron density and the drift velocity v_d:

$$J = -en_s v_d \tag{1.7.1}$$

In low temperature, the current is contributed mainly by electrons near the Fermi energy E_f. An electric field makes the entire distribution of states in k-space shift by $\boldsymbol{k}_d$

$$[f(\boldsymbol{k})]_{\boldsymbol{E}\neq 0} = [f(\boldsymbol{k} - \boldsymbol{k}_d)]_{\boldsymbol{E}=0} \tag{1.7.2}$$

where

$$\frac{\hbar \boldsymbol{k}_d}{m} = \boldsymbol{v}_d \Rightarrow \boldsymbol{k}_d = e\boldsymbol{E}\tau_m/\hbar \tag{1.7.3}$$

Assuming that the shit k_d is small compared to k_f, from a collective point view the electric field only moves a few electrons from $-k_f$ to $+k_f$, We could rewrite the current density in slightly different form to reflect this point view:

$$\boldsymbol{J} = e\left[n_s\frac{v_d}{v_f}\right]\boldsymbol{v_f} \tag{1.7.4}$$

1.7.1 Quasi-Fermi level separation

Quasi-Fermi level F^+ is defined for the electrons that move in the same direction as the force $e\boldsymbol{E}$ and another quasi-Fermi level F^- for the electrons that move in the opposite direction. All the states completely full and carry no current. But in the energy range between F^+ and F^-, the states in the $-k_x$ are empty while those in the $+k_x$ are full and it is the electrons in these states that carry the current. We can estimate F^+ and F^- by noting that

$$F^+ \sim \frac{\hbar^2(k_f + k_d)^2}{2m}, \quad F^- \frac{\hbar^2(k_f - k_d)^2}{2m} \tag{1.7.5}$$

Assuming that $k_f \gg k_d$, we have

$$F^+ - F^- \sim \frac{\hbar^2 k_f}{m}2k_d = 2eEv_f\tau_m = 2eEL_m \tag{1.7.6}$$

The separation of quasi-Fermi levels is propotional to the energy that an electron gains in electric field over a mean free path.

1.7.2 Einstein relation

Consider a situation where a bias voltage V is applied across a rectangular conductor with length L and width W. The electric field in the conductor is given by $E = \hat{x}V/L$ in the conductor. The buttom of the subband E_s acquaires a constant slope proportional to the electric field:

$$\boldsymbol{E} = \nabla E_s/e \tag{1.7.7}$$

In a homogeneous conductor, the electron density is constant everywhere. Since the electron density is a function of $(F_n - E_s)$, this means that the quasi-Fermi level $F_n = (F^+ + F^-)/2$ and the band-edge E_s run parallel to each other. The potential in contact 1 is μ_1 and that in contact 2 is μ_2. In this energy range, the electron density is $N_s(\mu_1 - \mu_2)$ in contact 1 and no electrons in contact 2. Since there is a concentration gradient from contact 1 to contact 2. This sets up a diffusion current can be written from diffusion equation as:

$$\boldsymbol{J} = -eD\nabla n = e^2 DN_s \frac{(\mu_1 - \mu_2)}{eL}\hat{x} = e^2 DN_s \boldsymbol{E} \tag{1.7.8}$$

Comparing with the relation $\boldsymbol{J} = \sigma\boldsymbol{E}$, we obtain the Einstein relation:

$$\sigma = e^2 N_s D \tag{1.7.9}$$

1.7.3 Drift or diffusion?

In the view of drift, the current comes from the entire sea of electrons, so that the conductivty in terms of mobility is given by

$$\sigma = en_s\mu, \quad \mu = \frac{v_d}{E} = \frac{e\tau_m}{m} \tag{1.7.10}$$

In the view of diffusion, the current comes from the electrons in the energy range between μ_1 and μ_2. The conductivity in terms of diffuction coefficient is given before. Combining the two expressions for conductivity, we have

$$en_s\mu = e^2 N_s D \Rightarrow \frac{eD}{\mu} = \frac{n_s}{N_s} = E_f - E_s \Rightarrow D = \frac{1}{2}v_f^2\tau_m \tag{1.7.11}$$

The Einstein relation in this case is obtained by replacing $(E_f - E_s)$ with k_BT for non-degenerate conductors.

$$\frac{eD}{\mu} = k_Bt \tag{1.7.12}$$

Chapter 2

Conductance from transmission

2.1 Resistance of a ballistic conductor

Consider a piece conductor connected to two large contact pads. If the dimensions of the conductor are large then its conductance would be given by $G = \sigma W/L$. If the length L of the conductor is reduced to a value comparable to or smaller than the mean free path l, then the conductor becomes ballistic and its conductance is no longer given by the above formula. In a ballistic conductor, the resistance arising from the interface between the conductor, is called contact resistance(G_c^{-1}).

2.2 Reflectionless contacts

The contact are reflectionless that the electrons can enter it from the conductor without suffering reflection. In reflectionless contacts, a simple situation is that $+k$ states in the conductor are occupied only by electrons originating in the left contact while $-k$ states are occupied only by electrons originating in the right contact. This is because the electrons in the left contact have positive velocities and can only enter the conductor in $+k$ states, while those in the right contact have negative velocities and can only enter the conductor in $-k$ states. The quasi-Fermi level F^+ for the $+k$ states is always equal to μ_1 while that for the $-k$ states F^- is always equal to μ_2.

2.2.1 Calculating the current

Each transverse mode has a dispersion relation $E(N, k)$ with a cut-off energy $\epsilon_N = E(N, k = 0)$, The number of transverse mode at energy E is obtained by counting the number of modes having cut-off energies less than E:

$$M(E) = \sum_N \theta(E - \epsilon_N) \tag{2.2.1}$$

Consider a single transverse mode whose $+k$ states are occupied according to $f^+(E)$. The current carried by this mode is given by

$$I_N^+ = \frac{e}{L} \sum_k v f^+(E) = \frac{e}{L} \sum_k \frac{1}{\hbar} \frac{\partial E(N, k)}{\partial k} f^+(E) \tag{2.2.2}$$

Assuming periodic boundary conditions and converting the sum over k into an integral according to the usual prescription

$$\sum_k \to g_s \frac{L}{2\pi} \int \mathrm{d}k \tag{2.2.3}$$

we have

$$I_N^+ = \frac{2e}{h}\int \frac{\partial E(N,k)}{\partial k} f^+(E)\mathrm{d}k = \frac{2e}{h}\int f^+(E)\mathrm{d}E \tag{2.2.4}$$

Extend this to all the transverse modes, we have

$$I^+ = \frac{2e}{h}\int M(E) f^+(E)\mathrm{d}E \tag{2.2.5}$$

2.2.2 Contact resistance

Assuming that the number of modes M is constant over the energy range $\mu_1 > E > \mu_2$, the total current is given by

$$I = I^+ - I^- = \frac{2e^2}{h} M \frac{(\mu_1 - \mu_2)}{|e|} \Rightarrow G_c = \frac{2e^2}{h} M \tag{2.2.6}$$

Note that the contact conductance is dependent only on the number of transverse modes M at the Fermi energy. The number of transverse modes M can be estimated by periodic boundary conditions. The allowed values of k_y are spaced by $2\pi/W$, each value of k_y corresponding to a distinct transverse mode. Thus the total number of transverse modes at the Fermi energy is given by

$$M = Int\left[\frac{2k_f}{2\pi/W}\right] = Int\left[\frac{W}{\lambda_f/2}\right] \tag{2.2.7}$$

where $Int[x]$ denotes the integer part of x.

2.3 Landauer formula

The Landauer formula relates the conductance of a conductor to its transmission properties T. For a conductor connected to two contacts, the conductance G is given by

$$G = \frac{e^2}{\pi\hbar} MT \tag{2.3.1}$$

The influx of electrons from contact 1 is given by

$$I_1^+ = (2e/h)M[\mu_1 - \mu_2] \tag{2.3.2}$$

The outflux of electrons into contact 2 is given by

$$I_2^+ = (2e/h)MT[\mu_2 - \mu_1] \tag{2.3.3}$$

The rest of the flux is reflected back to contact 1:

$$I_1^- = (2e/h)M(1-T)[\mu_1 - \mu_2] \tag{2.3.4}$$

The net current flowing at any point in the device is given by

$$I = I_1^+ - I_1^- = I_2^+ = \frac{e^2}{\pi\hbar} MT \frac{(\mu_1 - \mu_2)}{|e|} \Rightarrow G = \frac{e^2}{\pi\hbar} MT \tag{2.3.5}$$

The Landauer formula can be viewed as a mesoscopic version of the Einstein relation:

$$G = e^2 N_s D \frac{W}{L} \rightleftarrows G = \frac{e^2}{\pi\hbar} MT \tag{2.3.6}$$

with the conductivity replaced by the conductance, the density of states replaced by the number of transverse modes, and the diffusion constant replaced by the transmission probability:

$$\sigma \rightleftarrows G, \quad N_s \rightleftarrows M, \quad D \rightleftarrows T \tag{2.3.7}$$

2.3.1 Ohm's law

The Landauer formula incorporates two properties of the resistance of small conductors, which are the length-independnt interface resistance with the contacts and the discrete steps related to the transverse modes in narrow conductors.For a wide conductor with width W, the number of transverse modes is $M \sim k_f W/\pi$, the conductance can be written as

$$G = \frac{e^2}{\pi\hbar} MT = \frac{e^2}{\pi\hbar}\frac{k_f W}{\pi} T \tag{2.3.8}$$

The transmission probability T is given by

$$T = \frac{L_0}{L + L_0} \tag{2.3.9}$$

where L_0 is a characteristic length related to the mean free path.

$$G = \frac{W}{L + L_0} e^2 N_s (v_f L_0/\pi) \tag{2.3.10}$$

Identifying the diffusion coefficient as $D = v_f L_0/\pi$ and make use of the Einstein relation, we have

$$G = \frac{\sigma W}{L + L_0} \Rightarrow G^{-1} = \frac{L + L_0}{\sigma W} \tag{2.3.11}$$

We could write this as a series combination of the contact resistance and the interface resistance, which obeys Ohm's law:

$$G_s^{-1} = \frac{L}{\sigma W}, \quad G_c^{-1} = \frac{L_0}{\sigma W} \tag{2.3.12}$$

2.3.2 Transmission probability

The transmission probability from a to b is T_{ab}, while the transmission probability around a and b are T_a, T_b, and the reflection probability $R_a = 1 - T_a, R_b = 1 - T_b$. We can write

$$T_{ab} = T_a T_b + T_a T_b R_a R_b + T_a T_b (R_a R_b)^2 + ... = \frac{T_a T_b}{1 - R_a R_b} \tag{2.3.13}$$

We can rewrite this in the form

$$\frac{1 - T_{ab}}{T_{ab}} = \frac{1 - T_a}{T_a} + \frac{1 - T_b}{T_b} \tag{2.3.14}$$

The transmission probability $T(N)$ of N scatters in series, each having a transmission probability T is given by

$$\frac{1 - T(N)}{T(N)} = N\frac{1 - T}{T} \Rightarrow T(N) = \frac{T}{N(1 - T) + T} \tag{2.3.15}$$

The number of scatters N in a conductor of length L can be written as νL where ν is the linear density of scatters. Hence we can write

$$T(L) = \frac{T}{\nu L(1 - T) + T} = \frac{L_0}{L + L_0}, \quad L_0 = \frac{T}{\nu(1 - T)} \tag{2.3.16}$$

It is easy to see that the length L_0 is of the order of a mean free path. A mean free path is the average distance an electron can travel before it is scattering. Since the probability of scattering by an individual scatter is $(1 - T)$ we can write(assuming $T \sim 1$)

$$(1 - T)\mu L_m \sim 1 \rightarrow L_m \sim \frac{1}{\mu(1 - T)} \sim L_0 \tag{2.3.17}$$

2.3.3 Energy distribution of electrons

At finite temperatures, the distribution fucntion $f^+(E)$ for the $+k$ states is given by(left of scatter)

$$f^+(E) = \theta(\mu_1 - E) \tag{2.3.18}$$

The distribution function $f^-(E)$ for the $-k$ states is given by(right of scatter)

$$f^-(E) = \theta(\mu_2 - E) \tag{2.3.19}$$

The situation is a little more complicated when we consider the occupation of the $-k$ states before the scatter or the $+k$ state after the scatter. All states are completely filled below μ_2. But in the energy range between μ_1 and μ_2, the $+k$ states to the right of the scatter are filled only partially(with probability T) via transmitted electrons so that(near right)

$$f^+(E) = \theta(\mu_2 - E) + T\left[\theta(\mu_1 - E) - \theta(\mu_2 - E)\right] \tag{2.3.20}$$

Similarly, the $-k$ states to the left of the scatter are filled only partially(with probability $1-T$) via reflected electrons so that(near left)

$$f^-(E) = \theta(\mu_2 - E) + (1-T)\left[\theta(\mu_1 - E) - \theta(\mu_2 - E)\right] \tag{2.3.21}$$

Moving more than a few energy relaxation lengths away from the scatter, the electrons will settle down to lower energies and a Fermi distribution will be established
(far left)

$$f^-(E) = \theta(F' - E) \tag{2.3.22}$$

(far right)

$$f^+(E) = \theta(F'' - E) \tag{2.3.23}$$

The quasi-Fermi levels F' and F'' can be found by noting that the total number of electrons must be conserved. Equating the total number of electrons on the left side before and after the scatter, we have

$$F' = \mu_2 + (1-T)(\mu_1 - \mu_2) \tag{2.3.24}$$

Similarly, equating the total number of electrons on the right side before and after the scatter, we have

$$F'' = \mu_2 + T(\mu_1 - \mu_2) \tag{2.3.25}$$

2.3.4 Spational variation of the electrochemical potential

It is convenient to defien normalized potential μ^+ and μ^- which are obtained by F^+ and F^- by setiting $\mu_2 = 0$ and $\mu_1 = 1V$:

$$\mu^+ = 1 \quad \text{(left)}, \quad \mu^+ = T \quad (right) \tag{2.3.26}$$

Similarly the normalized potential for the $-k$ states is given by

$$\mu^- = 1 - T \quad \text{(left)}, \quad \mu^- = 0 \quad (right) \tag{2.3.27}$$

There is a sharp drop across the scatter just expected for a localized resistance. The nomalized potential drop across the scatter is equal to $(1-T)$ for both the positive and negative k-state. The actual potential drop eV_s is simply the normalized drop multiplied by the total applied voltage $\mu_1 - \mu_2$:

$$eV_s = (1-T)(\mu_1 - \mu_2) \tag{2.3.28}$$

The rest if the applied bias voltage $T(\mu_1 - \mu_2)$ is dropped at the interfaces with the contacts. These are prescisely the potential drops we would expect for a current

$$I = \frac{e^2}{\pi\hbar} MT \frac{(\mu_1 - \mu_2)}{|e|} \tag{2.3.29}$$

flowing through the scatterer resistance in series with the contact resistance:

$$G_s^{-1} = \frac{\pi\hbar}{e^2 M}\frac{1-T}{T}, \quad G_c^{-1} = \frac{\pi\hbar}{e^2 M}(1-T) \tag{2.3.30}$$

2.3.5 heat dissipation

The evolution of the electrorn energy distribution requires the dissipation of heat and it takes place over a distance of the order of an energy relaxatio length from the scatter. The heat dissipation P_D is given by the spatial gradient of the energy current I_U:

$$P_D = \frac{\mathrm{d}}{\mathrm{d}z} I_U \tag{2.3.31}$$

The energy current I_U can be calculated from the energy disdribution iE of the current that flows at any spatial location.

$$I_U = \frac{1}{e}\int Ei(E)\mathrm{d}E \tag{2.3.32}$$

Noting that the net current I is given by

$$I = \frac{1}{e}\int i(E)\mathrm{d}E \tag{2.3.33}$$

we could define an average energy U of the current as

$$U = \frac{I_U}{I} = \frac{\int Ei(E)\mathrm{d}E}{\int i(E)\mathrm{d}E} \tag{2.3.34}$$

The heat dissipation can be written as

$$P_D = I\frac{\mathrm{d}U}{\mathrm{d}z} \tag{2.3.35}$$

The current per unit energy given by

$$i(E) = \frac{2eM}{h}\left[f^+(E) - f^-(E)\right] \tag{2.3.36}$$

2.4 Buttiker formula

The Buttiker formula is a generalization of the Landauer formula to multi-terminal conductors. Consider a conductor connected to several contacts. The current I_p flowing into contact p is extended from the two-terminal linear response formula

$$I = \frac{2e}{h}\overline{T}\left[\mu_1 - \mu_2\right] \tag{2.4.1}$$

by summing over all the terminals (indexed by p, q) as flows

$$I_p = \frac{2e}{h}\sum_q \left[\overline{T}_{q\leftarrow p}\mu_p - \overline{T}_{p\leftarrow q}\mu_q\right] \tag{2.4.2}$$

We can rewrite this in the form (with $V = \mu/e$)

$$I_p = \sum_q [G_{qp}V_p - G_{pq}V_q], \quad G_{pq} = \frac{2e^2}{h}\overline{T}_{p\leftarrow q} \tag{2.4.3}$$

In order to ensure that the current is zero when all the potentials are equal:

$$\sum_q G_{pq} = \sum_q G_{qp} \tag{2.4.4}$$

Thus the equation of I_p can be written as

$$I_p = \sum_q G_{pq}(V_p - V_q) \tag{2.4.5}$$

That beacuse the current flowing into contact p is the sum of the currents flowing between contact p and all other contacts q. The conductance coefficient (G) also obeys the relation which can be proved for coherent transport using properties of the S-matrix in next Chapter.(B is the magnetic field)

$$[G_{qp}]_{+B} = [G_{pq}]_{-B} \tag{2.4.6}$$

The potential at a voltage probe can be written as (setting $I_p = 0$)

$$V_p = \frac{\sum_{q\neq p} G_{pq}V_q}{\sum_{q\neq p} G_{pq}} \tag{2.4.7}$$

2.4.1 Three-terminal conductor

Consider a three-terminal conductor with contacts 1, 2, and 3. To calculate the resistance $R_{3t} = V/I$

$$\begin{pmatrix} I_1 \\ I_2 \\ I_3 \end{pmatrix} = \begin{bmatrix} G_{12}+G_{13} & -G_{12} & -G_{13} \\ -G_{21} & G_{21}+G_{23} & -G_{23} \\ -G_{31} & -G_{32} & G_{31}+G_{32} \end{bmatrix} \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} \tag{2.4.8}$$

The sum rules ensure that the Kirchhoff's current law is satisfied. Setting $V_3 = 0$ and truncating the third row and third column, we have

$$\begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{bmatrix} G_{12}+G_{13} & -G_{12} \\ -G_{21} & G_{21}+G_{23} \end{bmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix} \tag{2.4.9}$$

Inverting the matrix, we have

$$\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix}, \quad \begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix} = \begin{bmatrix} G_{12}+G_{13} & -G_{12} \\ -G_{21} & G_{21}+G_{23} \end{bmatrix}^{-1} \tag{2.4.10}$$

The resistance R_{3t} measured in the configureation shown in Fig.2.4.2a is given by

$$R_{3t} = \left[\frac{V_2}{I_1}\right]_{I_2=0} = R_{21} \tag{2.4.11}$$

The resistance R_{3t} measured in the configureation shown in Fig.2.4.2b is given by

$$R'_{3t} = \left[\frac{V_1}{I_2}\right]_{I_1=0} = R_{12} \tag{2.4.12}$$

2.4.2 Four-terminal conductor

Setting the voltage at one of the terminals (V_4) equal to zero, we have

$$\begin{pmatrix} I_1 \\ I_2 \\ I_3 \end{pmatrix} \begin{bmatrix} G_{12}+G_{13}+G_{14} & -G_{12} & -G_{13} \\ -G_{21} & G_{21}+G_{23}+G_{24} & -G_{23} \\ -G_{31} & -G_{32} & G_{31}+G_{32}+G_{34} \end{bmatrix} \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} \tag{2.4.13}$$

Inverting the matrix, we have

$$\begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix} \begin{pmatrix} I_1 \\ I_2 \\ I_3 \end{pmatrix} \tag{2.4.14}$$

$$\begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix} = \begin{bmatrix} G_{12}+G_{13}+G_{14} & -G_{12} & -G_{13} \\ -G_{21} & G_{21}+G_{23}+G_{24} & -G_{23} \\ -G_{31} & -G_{32} & G_{31}+G_{32}+G_{34} \end{bmatrix}^{-1} \tag{2.4.15}$$

The resistance R_{4t} measured in the configureation shown in Fig.2.4.3a is given by

$$R_{4t} = \frac{V}{I} = \left[\frac{V_2 - V_3}{I_1}\right]_{I_2=I_3=0} = R_{21} - R_{31} \tag{2.4.16}$$

while the resistance R'_{4t} measured in the configureation shown in Fig.2.4.3b is given by

$$R'_{4t} = \frac{V'}{I'} = \left[\frac{V_1}{I_2}\right]_{I_1=0, I_2=-I_3} = R_{12} - R_{13} \tag{2.4.17}$$

2.5 Non-zero temperature and bias

In zero temperature, we assume that the current was carried by a singel energy channel around the Fermi energy. This allowed us to write the current as

$$I = \frac{2e}{h}\overline{T}(\mu_1 - \mu_2) \tag{2.5.1}$$

where $\overline{T}$ denotes the product of the number of modes M and the transmission probability per mode T at the Fermi energy (assumed constant over the range μ_1 to μ_2). However, in finite temperatures, the electron energy distribution is broadened over an energy range of several $k_B T$ around the Fermi energy. The influx of electrons per unit energy from lead 1 is given by

$$i_1^+(E) = \frac{2e}{h} M f_1(E) \tag{2.5.2}$$

while the influx from lead 2 is given by (M' is the number of modes in lead 2)

$$i_2^-(E) = \frac{2e}{h} M' f_2(E) \tag{2.5.3}$$

The outflux from lead 2 is written as

$$i_2^+(E) = T i_1^+(E) + (1 - T') i_2^-(E) \tag{2.5.4}$$

while the outflux from lead 1 is written as

$$i_1^-(E) = (1 - T) i_1^+(E) + T' i_2^-(E) \tag{2.5.5}$$

The net current $i(E)$ flowing at any point in the device is given by

$$\begin{aligned} i(E) &= i_1^+(E) - i_1^-(E) \\ &= i_2^+(E) - i_2^-(E) \\ &= T i_1^+(E) - T' i_2^-(E) \\ &= \frac{2e}{h}\left[T(E)M(E)f_1(E) - T'(E)M'(E)f_2(E)\right] \end{aligned} \tag{2.5.6}$$

Defining the transmission function as $\overline{T}(E) = T(E)M(E)$ we can write

$$i(E) = \frac{2e}{h}\left[\overline{T}(E)f_1(E) - \overline{T}'(E)f_2(E)\right] \tag{2.5.7}$$

The total current I is obtained by integrating over all energies:

$$\begin{aligned} I &= \int i(E)\mathrm{d}E \\ &= \frac{2e}{h}\int\left[\overline{T}(E)f_1(E) - \overline{T}'(E)f_2(E)\right]\mathrm{d}E \\ &= \frac{2e}{h}\int \overline{T}(E)\left[f_1(E) - f_2(E)\right]\mathrm{d}E \end{aligned} \tag{2.5.8}$$

Assuming that there is no inelastic scattering inside the device, then it can be shown that $\overline{T}(E) = \overline{T}'(E)$.

2.5.1 Linear response

For small deviations from equilibrium, the current is proportional to the applied bias voltage.

$$\delta I = \frac{2e}{h}\int\left(\left[\overline{T}(E)\right]_{eq}\delta[f_1 - f_2] + [f_1 - f_2]_{eq}\delta\left[\overline{T}(E)\right]\right)\mathrm{d}E \tag{2.5.9}$$

The second term is clearly zero. We can simplify the first term by using a Taylor's series expansion to write

$$\delta[f_1 - f_2] = [\mu_1 - \mu_2]\left(-\frac{\partial f_0}{\partial E}\right)_{eq} \tag{2.5.10}$$

where $f_0(E)$ is the equilibrium Fermi distribution function.

$$f_0(E) = \left[\frac{1}{\exp\left[(E-\nu)/k_BT\right]+1}\right]_{\mu=E_f} \tag{2.5.11}$$

Thus we have

$$G = \frac{\delta I}{(\mu_1 - \mu_2)/e} = \frac{2e^2}{h}\int\left(-\frac{\partial f_0}{\partial E}\right)_{eq}\left[\overline{T}(E)\right]_{eq}\mathrm{d}E \tag{2.5.12}$$

At low temperatures, the derivative of the Fermi function approaches a delta function centered at the Fermi energy:

$$f_0(E) \approx \theta(E_f - E) \Rightarrow -\frac{\partial f_0}{\partial E} \approx \delta(E - E_f) \tag{2.5.13}$$

Thus we have(drop the subscript 'eq' for clarity)

$$G = \frac{2e^2}{h}\overline{T}(E_f) \tag{2.5.14}$$

2.5.2 When is the response linear?

The response is linear regardless of the temperature if the transmission functin $\overline{T}(E)$ is approximately constant over over the energy range where transport occurs, and can be assumed to unaffeected by the applied bias voltage. Thus we have

$$I = \frac{2e}{h}\overline{T}\int [f_1(E) - f_2(E)]\,\mathrm{d}E \tag{2.5.15}$$

at low temperatures it is easy to see that

$$\int [f_1(E) - f_2(E)]\,\mathrm{d}E = \mu_1 - \mu_2 \tag{2.5.16}$$

since $f_1(E) \approx \theta(\mu_1 - E)$ and $f_2(E) \approx \theta(\mu_2 - E)$. Thus we obtain a linear relationship between the current and the applied bias voltage:

$$I = \frac{2e}{h}\overline{T}(\mu_1 - \mu_2) \tag{2.5.17}$$

for arbitrary temperatures, as long as the transimission function is independent of energy and unaffected by the applied bias voltage. We can arrive a general criterion for linear response by rewriting the current as

$$I = \frac{1}{e}\int_{\mu_2}^{\mu_1} \hat{G}(E')\mathrm{d}E' \tag{2.5.18}$$

where the conductance function is defined as

$$\hat{G}(E) = \frac{2e^2}{h}\int \overline{T}(E)F_T(E - E')dE \tag{2.5.19}$$

$F_T(E)$ being the thermal broadening function

$$F_T(E) = -\frac{\mathrm{d}}{\mathrm{d}E}\left[\frac{1}{\exp(E/k_BT) + 1}\right] = \frac{1}{4k_BT}\operatorname{sech}^2\left(\frac{E}{2k_BT}\right) \tag{2.5.20}$$

The current is the linear response of the bias voltage if the conductance function $\hat{G}(E)$ is approximately constant over the energy range μ_1 to μ_2. Thus the current is wrriten as

$$I = \hat{G}(E_f)\frac{(\mu_1 - \mu_2)}{e} \tag{2.5.21}$$

where the conductance is given by

$$G = \hat{G}(E_f) = \frac{2e^2}{h}\int \overline{T}(E)F_T(E - E_f)\mathrm{d}E \tag{2.5.22}$$

For the transimssion function $\overline{T}(E)$ often changes rapidly with energy in low temperatures, the response is linear as long as the bias is much less than k_BT.

2.5.3 Multi-terminal conductors

For multi-terminal devices, the current can be written as

$$I_p = \int i_p(E)\mathrm{d}E, \quad i_p(E) = \frac{2e}{h}\sum_q \left[\overline{T}_{qp}(E)f_p(E) - \overline{T}_{pq}(E)f_q(E)\right] \tag{2.5.23}$$

Since there can be no current flowing at equilibrium, $f_p(E) = f_q(E) = f_0(E)$, we must have $\sum_q \overline{T}_{qp}(E) = \sum_q \overline{T}_{pq}(E)$. Thus we can rewrite the current as

$$I_p = \frac{2e}{h}\sum_q \overline{T}_{pq}(E)\,[f_p(E) - f_q(E)] \tag{2.5.24}$$

A floating current, i.e., $I_p = \int i_p(E)\mathrm{d}E = 0$. We can linearize the current for small deviations from equilibrium

$$I_p = \sum_q G_{pq}(V_p - V_q), \quad G_{pq} = \frac{2e^2}{h}\int \overline{T}_{pq}(E)\left(-\frac{\partial f_0}{\partial E}\right)\mathrm{d}E \tag{2.5.25}$$

at low temperatures, we have

$$G_{pq} = \frac{2e^2}{h}\overline{T}_{pq}(E_f) \tag{2.5.26}$$

2.6 Exclusion principle?

To modify the current expression to account for the exclusion principle

$$\begin{aligned} i_p(E) &= \frac{2e}{h}\sum_q \left[\overline{T}_{qp}(E)f_p(E)[1-f_q(E)] - \overline{T}_{pq}(E)f_q(E)[1-f_p(E)]\right] \\ &= \frac{2e}{h}\sum_q \left(\left[\overline{T}_{qp}(E)f_p(E) - \overline{T}_{pq}(E)f_q(E)\right] - \left[\overline{T}_{qp}(E) - \overline{T}_{pq}(E)\right] f_q(E)f_p(E)\right) \end{aligned} \tag{2.6.1}$$

2.6.1 Scattering states

Instinctively, we tend to think of the current as arising from the electronic transitions between an eigenstate localized in lead p and an eigenstate localized in q

$$|q,k\rangle \equiv \sin(kx_p) \to |p,k'\rangle \equiv \sin(k'x_q) \tag{2.6.2}$$

The wavefunction in lead p due to a scattering state (q,k) is given by ($\delta_{pq} = 1$ if $p = q$, zero otherwise)

$$\Psi_p(q) = \delta_{pq}\chi_p^+(y_p)\exp\left(ik^+x_p\right) + s'_{pq}\chi_p^-(y_p)\exp\left(ik^-x_p\right) \tag{2.6.3}$$

where χ_p^+ and χ_p^- are the transverse wavefunctions for the incident and scattered waves respectively. If the vector potential is zero in the lead then the two fucntions are identical and $k^- = -k^+$. A scatter state (q,k) gives rise to a current $i_p(q)$ per unit energy in lead p which is given by

$$i_p(q) = \frac{2e}{h}(\delta_{pq} - T_{pq}) \tag{2.6.4}$$

Hence a scattering state (q,k) occupied with probability $f_q(E)$ gives rise to a current in terminal p can be written as

$$\begin{aligned} I_p &= \int \sum_q f_q(E) i_p(q)\mathrm{d}E \\ &= \frac{2e}{h}\int \sum_q f_q(E)\left(\delta_{pq} - T_{pq}\right)\mathrm{d}E \end{aligned} \tag{2.6.5}$$

For coherent transport, the transimission coefficient obey the sum rule

$$\sum_q T_{pq} = \sum_q T_{qp} = 1 \tag{2.6.6}$$

Thus we can rewrite the current as

$$I_p = \frac{2e}{h}\int \sum_q T_{pq}(f_p(E) - f_q(E))\mathrm{d}E \tag{2.6.7}$$

Consider multiple modes in each lead, we obtain the same expression with the transimission coefficient replaced by the transmission function $\overline{T}_{pq}(E)$, by summing over all the modes in each lead.

$$\overline{T}_{pq}(E) = \sum_{m\in p}\sum_{n\in q} T_{mn} \tag{2.6.8}$$

2.6.2 Can scattering states be defined in non-zero magnetic fields?

In the presence of magnetic field, We need to choose our gauge such that the vector potential has the form

$$\boldsymbol{A} \sim \hat{x}_q B y_q \tag{2.6.9}$$

in every lead q. It has been shown that this can be done even for conductors with multiple leads arranged arbitrarily.

2.6.3 Are different transverse modes independent?

The general expressioin for calculating the current density J from the wavefunction Ψ is given by

$$\boldsymbol{J} = \frac{e}{2m}\left(\Psi(\hat{\boldsymbol{p}} - e\boldsymbol{A})\Psi^* + \Psi^*(\hat{\boldsymbol{p}} - e\boldsymbol{A})\Psi\right) \tag{2.6.10}$$

To obtain the total current carried by a lead we integrate the longitudinal(x-directed) current over the transverse coordinate (y):

$$I = \int \mathrm{d}y J_x(y) = \frac{e}{2m}\int \left(\Psi(\hat{p}_x - eA_x)\Psi^* + \Psi^*(\hat{p}_x - eA_x)\Psi\right)\mathrm{d}y \tag{2.6.11}$$

If the wavefunction consists of an incident wave

$$\Psi_i = \frac{1}{\sqrt{L}}\chi^+(y)\exp\left(ik^+x\right) \tag{2.6.12}$$

then the current is given by (assuming $\chi^+(y)$ is real)

$$I_i = \frac{e}{mL}\int \chi^+(y)(\hbar k^+ - eA_x)\chi^+(y)\mathrm{d}y \tag{2.6.13}$$

On the other hand, if the wavefunction consists of only a scattered wave of amplitude s'

$$\Psi_s = \frac{s'}{\sqrt{L}}\chi^-(y)\exp(ik^-x) \tag{2.6.14}$$

then the current is given by

$$I_s = \frac{e|s'|^2}{mL}\int \chi^-(y)(\hbar k^- - eA_x)\chi^-(y)\mathrm{d}y \tag{2.6.15}$$

But if the wavefunction consists of both incident and scattered waves

$$\Psi = \frac{1}{\sqrt{L}}\left[\chi^+(y)\exp\left(ik^+x\right) + s'\chi^-(y)\exp(ik^-x)\right] \tag{2.6.16}$$

Assuming that the current is given by $I = I_i - I_s$, a direct evaluation of the current would give rise to cross-terms of the form

$$\frac{e}{mL}\int\left[\chi^+\left(\frac{\hbar(k^+ + k^-)}{2} - eA_x\right)\chi^-\right]\mathrm{d}y \tag{2.6.17}$$

These cross-terms vanish by virtue of the following orthogonality relation between the transverse wavefunctions:

$$\int \chi^+(y)\left(\frac{\hbar(k + k')}{2} - eA_x\right)\chi^-(y)\mathrm{d}y = 0 \tag{2.6.18}$$

To proof this, we start from the transverse Schrodinger equation for the two modes:

$$\left[\frac{1}{2m}\left(\hbar k^\pm - eA_x\right)^2 + V(y)\right]\chi^\pm(y) = E_t^\pm\chi^\pm(y) \tag{2.6.19}$$

Multiplying the equation for χ^+ by χ^- and the equation for χ^- by χ^+, subtracting the two resulting equations, and integrating over y, we obtain

$$(E_t^+ - E_t^-)\int \chi^+(y)\chi^-(y)\mathrm{d}y = \frac{\hbar}{m}(k^+ - k^-)\int \chi^+(y)\left(\frac{\hbar(k^+ + k^-)}{2} - eA_x\right)\chi^-(y)\mathrm{d}y \quad (2.6.20)$$

The left-hand side is zero because of the orthogonality of the transverse wavefunctions corresponding to different eigenvalues. The right-hand side is zero unless $k^+ = k^-$. Thus we have proved the orthogonality relation.

2.6.4 Non-coherent transport

The key insight is an observation due to Buttiker that a voltage probe acts as a phase-breaking scatterer. This can be understood by considering the current flow from terminal 1 to 2 in a device having a voltage probe in between as shown in Fig.2.6.2. We can calculate the S-matrix for such a device using the Schrodinger equation if we assume that transport is fully coherent between any pair of terminals. However, the voltage probe is adjusted such that no net current flows into or out of probe. This means that any electron entering probe from the device must be replaced by another electron injected back into the device from probe. Since there is no phase relationship between these two electrons, the effect is to randomize the phase of the electron wavefunction. Thus the voltage probe acts as a phase-breaking scatterer. We can reverse this argument to conclude that phase-breaking scatterers can be simulated by introducing conceptual voltage probes where none exist in the real structure.

A conceptual probe with the appropriate properties can thus be used to simulate the effect of phase-breaking processes. A partially coherent conductor or an incoherent conductor can be visualized as a fully coherent conductor with a conceptual probe stuck to it. The current in such a structure can be described by an equation of the form

$$i_p(E) = \frac{2e}{h}\sum_q \overline{T}_{pq}(E)\,[f_p(E) - f_q(E)] + \frac{2e}{h}\overline{T}_{p\phi}(E)\,[f_p(E) - f_\phi(E)] \quad (2.6.21)$$

where the index q runs over all the real terminals while the index ϕ denotes a fictitious phase-breaking probe attached to the coonductor. The current at the fictitious probe is given by

$$i_\phi(E) = \frac{2e}{h}\sum_q \overline{T}_{\phi q}(E)\,[f_\phi(E) - f_q(E)] \quad (2.6.22)$$

Express the distribution function $f_\phi(E)$ at the fictitious probe in terms of the distribution functions at the real terminals by setting $i_\phi(E) = 0$:

$$f_\phi(E) = \frac{1}{R_\phi}\left(i_\phi + \sum_q \overline{T}_{\phi q} f_q\right), \quad R_\phi \equiv \sum_q \overline{T}_{\phi q} \quad (2.6.23)$$

Substituting this into the expression for $i_p(E)$, we have

$$i_p(E) = \frac{2e}{h}\sum_q \overline{T}_{pq}^{eff}[f_p - f_q] - \frac{2e}{h}\left[\overline{T}_{p\phi}/\overline{R_\phi}\right] i_\phi \quad (2.6.24)$$

The net current of fictitious probe integrated over all energies is zero

$$\int i_\phi(E)\mathrm{d}E = 0 \quad (2.6.25)$$

2.7 Equilibrium Current Fluctuations

We have seen that the conductance is given by ($f_0(E)$ is the equilibrium Fermi distribution function)

$$\begin{aligned} G &= \frac{2e^2}{h}\int \overline{T}(E)\left(-\frac{\partial f_0}{\partial E}\right)\mathrm{d}E \\ &= \frac{2e^2}{hk_BT}\int \overline{T}(E)f_0(E)[1-f_0(E)]\mathrm{d}E \end{aligned} \tag{2.7.1}$$

We could rewrite this in the form

$$G = \frac{|e|I_{eq}}{k_BT} \tag{2.7.2}$$

where I_{eq} is defined as

$$I_{eq} = \frac{2|e|}{h}\int \overline{T}(E)f_0(E)[1-f_0(E)]\mathrm{d}E \tag{2.7.3}$$

In any conductor there are current fluctuations I_{eq} in both directions which balance out on the average at equilibrium. A small bias $\Delta\mu$ increases the flow in one direction to $I_{eq}+\Delta I$ such that

$$\frac{\Delta I}{I_{eq}} = \frac{\Delta\mu}{k_BT} \Rightarrow G = \frac{\Delta I}{\Delta\mu/|e|} = \frac{|e|I_{eq}}{k_BT} \tag{2.7.4}$$

This result seems to be quite general. An example of this is a p-n junction diode where the current-voltage characteristic is given by

$$I = I_{eq}\left[\exp\left(\frac{eV}{k_BT}\right)-1\right] \Rightarrow G = \left.\frac{\mathrm{d}I}{\mathrm{d}V}\right|_{V=0} = \frac{eI_{eq}}{k_BT} \tag{2.7.5}$$

which suggests an intereseting relationship between the Landauer formula and the Nyquist-Johnson formula which relates the conductance to the correlation fuction for the equilibrium noise current:

$$G = \frac{1}{2k_BT}\int_{-\infty}^{\infty}\langle I(t_0+t)I(t_0)\rangle_{eq}\mathrm{d}t \tag{2.7.6}$$

The angle-brakets denote averaging over times t_0, or equalently, averaging over an ensemble of systems at equilibrium, which are consistent only if

$$\int_{-\infty}^{\infty}\langle I(t_0+t)I(t_0)\rangle_{eq}\mathrm{d}t = 2|e|I_{eq} \tag{2.7.7}$$

which seems reasonable if we view the equilibrium noise as the shot noise due to independent uncorrelated equilibrium current I_{eq} flowing in either direction.

2.8 Summary

The basic results of the Landauer-Buttiker formalism for coherent transport can be summarized here for convenience:

$$I_p = \int i_p(E)\mathrm{d}E, \quad i_p(E) = \frac{2e}{h}\sum_q \overline{T}_{pq}(E)\left[f_p(E)-f_q(E)\right] \tag{2.8.1}$$

Here $f_p(E)$ is the Fermi distribution function in terminal P

$$f_p(E) = \frac{1}{\exp\left[(E-\mu_p)/k_BT\right]+1} \tag{2.8.2}$$

and $\overline{T}_{pq}(E)$ is the transmission function from terminal q to p. The transimission function obeys the sum rule

$$\sum_q \overline{T}_{pq}(E) = \sum_q \overline{T}_{qp}(E) \tag{2.8.3}$$

If the bias voltage is small such that

$$\Delta\mu = \epsilon_c + (\text{a few } k_B T) \tag{2.8.4}$$

then we can linearize the current to obtain

$$I_p = \sum_q G_{pq}(V_p - V_q), \quad G_{pq} = \frac{2e^2}{h} \int \overline{T}_{pq}(E) \left(-\frac{\partial f_0}{\partial E}\right) \mathrm{d}E \tag{2.8.5}$$

where $f_0(E)$ is the equilibrium Fermi distribution function. At low temperatures, the conductance coefficients reduce to

$$G_{pq} = \frac{2e^2}{h} \overline{T}_{pq}(E_f) \tag{2.8.6}$$

Chapter 3

Transmission function, S-matrix, and Green's functions

3.1 Transmission function and the S-matrix

A coherent conductor can be characteristized at each energy by an S-matrix which relates the outgoing wave amplitudes to the incoming wave amplitudes at the different leads. To be specific, if there is a total of three modes in the leads as shown in Fig.3.1.1, we can write

$$\begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{bmatrix} s_{11} & s_{12} & s_{13} \\ s_{21} & s_{22} & s_{23} \\ s_{31} & s_{32} & s_{33} \end{bmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \tag{3.1.1}$$

where a_i and b_i are the incoming and outgoing wave amplitudes in mode i respectively. At any given energy E, we will denote the number of propagating modes in lead p by $M_p(E)$. The total number of modes in each lead

$$M_T(E) = \sum_p M_p(E) \tag{3.1.2}$$

The scattering matrix is dimensions $M_T \times M_T$. We can calculate the S-matrix starting from the (effective mass) Schrodinger equation

$$\left[E_s + \frac{(\mathrm{i}\hbar\nabla + e\boldsymbol{A})^2}{2m} + U(x,y)\right]\Psi(x,y) = E\Psi(x,y) \tag{3.1.3}$$

The transmission probability T_{nm} is obtained by taking the squared magniitude of the corresponding element of the S-matrix:

$$T_{m\leftarrow n} = |s_{m\leftarrow n}|^2 \tag{3.1.4}$$

The transmission function $\overline{T}_{pq}(E)$ is obtained by summing over all modes from lead q to lead p:

$$\overline{T}_{p\leftarrow q}(E) = \sum_{m\in q}\sum_{n\in p} T_{n\leftarrow m} \tag{3.1.5}$$

3.1.1 Unitarity

In order ensure current conservation, the S-matrix must be unitary. We can write in matrix notation

$$\{b\} = [S]\{a\} \tag{3.1.6}$$

Where the matrix $[S]$ has dimensions $M_T \times M_T$, while M_T being the total number of modes in all the leads, and $\{a\}$ and $\{b\}$ are column vectors representing the incoming and outgoing wave amplitudes in the different modes in the leads. Assuming that the incoming and outgoing currents in a particular mode m are proportional to the squared magnitudes of the corresponding mode amplitudes a_m and b_m. Current conservation requires that the total incoming current equals the total outgoing current:

$$\sum_m |a_m|^2 = \sum_m |b_m|^2 \Rightarrow \{a\}^\dagger\{a\} = \{b\}^\dagger\{b\} \tag{3.1.7}$$

Since $\{b\} = [S]\{a\}$, we have

$$\{a\}^\dagger\{a\} = \{a\}^\dagger[S]^\dagger[S]\{a\} \tag{3.1.8}$$

This must hold for any arbitrary vector $\{a\}$, which implies that

$$[S]^\dagger[S] = [I] \tag{3.1.9}$$

So that in terms of the elements of the S-matrix, we have

$$\sum_{m=1}^{M_T} |s_{mn}|^2 = 1 = \sum_{m=1}^{M_T} |s_{nm}|^2 \tag{3.1.10}$$

3.1.2 An import point

The current associated with a scattered wave is proportional to the square of the wavefunction multiplied by the velocity. It is customary to define the S-matrix in terms of the 'current amplited' which is equal to the wave amplitude times the square root of the velocity. Instead we could define a matrix $[S']$ in terms of the wave amplitudes directly. The two matrices are related by

$$s'_{mn} = s_{mn}\sqrt{\frac{v_m}{v_n}} \tag{3.1.11}$$

But the matrix $[S']$ is not unitary. Using the unitarity of $[S]$, we can show that

$$\sum_q \overline{T}_{qp} = \sum_{n\in p}\sum_{m=1}^{M_T} T_{mn} = \sum_{n\in p}\sum_{m=1}^{M_T} |s_{mn}|^2 = \sum_{n\in p} 1 = M_p \tag{3.1.12}$$

$$\sum_q \overline{T}_{pq} = \sum_{n\in p}\sum_{m=1}^{M_T} T_{nm} = \sum_{n\in p}\sum_{m=1}^{M_T} |s_{nm}|^2 = \sum_{m\in p} 1 = M_p \tag{3.1.13}$$

Hence the following sum rule is always satisfied by the transimission function:

$$\sum_q \overline{T}_{pq}(E) = \sum_q \overline{T}_{qp}(E) = M_p(E) \tag{3.1.14}$$

where $M_p(E)$ is the number of propagating modes in lead p at energy E. For a device with N leads we can write down the transmission function $\overline{T}$ in the form of an $N \times N$ matrix. Assuming $N = 3$ (to be specific), we have

$$\begin{array}{ccccc} \overline{T}_{pq}(E): & q=1 & q=2 & q=3 & \\ p=1 & \overline{T}_{11} & \overline{T}_{12} & \overline{T}_{13} & SUM = M_1 \\ p=2 & \overline{T}_{21} & \overline{T}_{22} & \overline{T}_{23} & SUM = M_2 \\ p=3 & \overline{T}_{31} & \overline{T}_{32} & \overline{T}_{33} & SUM = M_3 \\ SUM = & M_1 & M_2 & M_3 & \end{array} \tag{3.1.15}$$

The sum rules requires that the elements in each row and each column add up to the number of modes for that terminal. It is interesting to note that the sum rules imply that for 2D devices the transmission function is always reciprocal, even if a magnetic field is present. That is, $\overline{T}_{12}(E) = \overline{T}_{21}(E)$. To see this, consider a two-terminal device.

$$\begin{array}{cccc} \overline{T}_{pq}(E): & q=1 & q=2 & \\ p=1 & \overline{T}_{11} & \overline{T}_{12} & SUM = M_1 \\ p=2 & \overline{T}_{21} & \overline{T}_{22} & SUM = M_2 \\ SUM = & M_1 & M_2 & \end{array} \tag{3.1.16}$$

Since $\overline{T}_{11} + \overline{T}_{12} = M_1 = \overline{T}_{21} + \overline{T}_{11}$, we must have $\overline{T}_{12} = \overline{T}_{21}$.

3.1.3 Sum rule for the conductance matrix

We can obtain the following sum rules for the conductance matrix:

$$\begin{aligned} \sum_q G_{pq} = \sum_q G_{qp} &= \frac{2e^2}{h} \int \left(-\frac{\partial f_0}{\partial E}\right) \sum_q \overline{T}_{pq}(E) \mathrm{d}E \\ &= \frac{2e^2}{h} \int \left(-\frac{\partial f_0}{\partial E}\right) M_p(E) \mathrm{d}E = \frac{2e^2}{h} M_p(E_f) \text{ at low temperatures} \end{aligned} \tag{3.1.17}$$

3.1.4 Reciprocity

For coherent transport, the symmetry properties of the S-matrix ensure that the reciprocity relation is satisfied. The basic property of the S-matrix that we will use is that reversing the magneti field transposes the S-matrix:

$$[S]_{+B} = [S]^t_{-B}, \quad \text{or} \quad s_{mn}(+B) = s_{nm}(-B) \tag{3.1.18}$$

To prove this, suppose we have solved the Schrodinger equation

$$\left[E_s + \frac{(\mathrm{i}\hbar\nabla + e\boldsymbol{A})^2}{2m} + U(x,y)\right] \Psi(x,y) = E\Psi(x,y) \tag{3.1.19}$$

To obtain the S-matrix connecting the outgoing amplitudes b to the incoming amplitudes a. If we take the complex conjugate of the Schrodinger equation, we obtain

$$\left[E_s + \frac{(\mathrm{i}\hbar\nabla - e\boldsymbol{A})^2}{2m} + U(x,y)\right] \Psi^*(x,y) = E\Psi^*(x,y) \tag{3.1.20}$$

and reverse the vector potential $\boldsymbol{A} \to -\boldsymbol{A}$ (which is equivalent to reversing the magnetic field $\boldsymbol{B} = \nabla \times \boldsymbol{A}$)

$$\left[E_s + \frac{(\mathrm{i}\hbar\nabla + e\boldsymbol{A})^2}{2m} + U(x,y)\right] \Psi^*(x,y) = E\Psi^*(x,y) \tag{3.1.21}$$

Thus $\Psi^*(x,y)$ is also a solution of the Schrodinger equation, which means

$$[\psi^*(x,y)]_{-B} = [\psi(x,y)]_{+B} \tag{3.1.22}$$

Taking the complex conjugate turns an incoming wave into an outgoing wave and vice versa (for current density $J = \frac{\hbar}{m}\mathrm{Im}(\psi^*\nabla\psi)$, complex conjugate turns $J \to -J$). So if

$$\{b\} = [S]_{+B}\{a\} \to \{b^*\} = [S]^*_{+B}\{a^*\} \tag{3.1.23}$$

then we must have

$$\{a^*\} = [S]_{-B}\{b^*\} \to \{b^*\} = [S]^{-1}_{-B}\{a^*\} \tag{3.1.24}$$

Hence

$$[S]^{-1}_{-B} = [S]^{*}_{+B} \tag{3.1.25}$$

with the unitarity of the S-matrix

$$[S]^{\dagger}_{-B} = [S]^{-1}_{-B} \tag{3.1.26}$$

we obtain

$$[S]^{*}_{+B} = [S]^{\dagger}_{-B} \Rightarrow [S]_{+B} = [S]^{t}_{-B} \tag{3.1.27}$$

Next we take the squared magnitude of both sides and sum over all modes m in lead p and over all modes n in lead q to obtain

$$\sum_{m\in p}\sum_{n\in q} |s_{mn}|^2_{+B} = \sum_{m\in p}\sum_{n\in q} |s_{nm}|^2_{-B} \tag{3.1.28}$$

which implies the reciprocity relation for the transmission function:

$$\overline{T}_{p\leftarrow q}(+B) = \overline{T}_{q\leftarrow p}(-B) \tag{3.1.29}$$

Thus the conductance matrix obeys the Onsager reciprocity relation:

$$[G_{pq}]_{+B} = [G_{qp}]_{-B} \tag{3.1.30}$$

The property of reciprocity holds only for small bias, unlike the sum rules discussed in the last subsection which hold irrespective of the bias. The reason is that we need to assume the electrostatic potential $U(x, y)$ is inside the conductor remains unchanged when the magnetic field is reversed.

3.2 Combining S-matrices

Consider a four-terminal Hall bridge, divided into two sections which have S-matrices $[S^{(1)}]$ and $[S^{(2)}]$ respectively. We can combine them to obtain the composite S-matrix.

$$s = s^{(1)} \otimes s^{(2)} \tag{3.2.1}$$

Let us write the individual S-matrices in the form

$$\begin{pmatrix} b_{13} \\ b_5 \end{pmatrix} = \begin{bmatrix} r^{(1)} & t'^{(1)} \\ t^{(1)} & r'^{(1)} \end{bmatrix} \begin{pmatrix} a_{13} \\ a_5 \end{pmatrix}, \quad \begin{pmatrix} a_5 \\ b_{24} \end{pmatrix} = \begin{bmatrix} r^{(2)} & t'^{(2)} \\ t^{(2)} & r'^{(2)} \end{bmatrix} \begin{pmatrix} b_5 \\ a_{24} \end{pmatrix} \tag{3.2.2}$$

where $a_{13} = (a_1, a_3)^t$, $b_{13} = (b_1, b_3)^t$, $a_{24} = (a_2, a_4)^t$, and $b_{24} = (b_2, b_4)^t$. The matrices r, r', t, and t' are the reflection and transmission sub-matrices. To obtain the composite S-matrix,. Note that at terminal 5 in Fig.3.2.1, the incoming amplitude a_5 for section 1 is equal to the outgoing amplitude b_5 for section 2. We can eliminate a_5 and b_5 to obtain

$$\begin{pmatrix} b_{13} \\ b_{24} \end{pmatrix} = \begin{bmatrix} r^{(1)} + t'^{(1)}[I - r^{(2)}r'^{(1)}]^{-1}r^{(2)}t^{(1)} & t'^{(1)}[I - r^{(2)}r'^{(1)}]^{-1}t'^{(2)} \\ t^{(2)}[I - r'^{(1)}r^{(2)}]^{-1}t^{(1)} & r'^{(2)} + t^{(2)}[I - r'^{(1)}r^{(2)}]^{-1}r'^{(1)}t'^{(2)} \end{bmatrix} \begin{pmatrix} a_{13} \\ a_{24} \end{pmatrix} \tag{3.2.3}$$

let us denote the composite S-matrix by

$$[S] = \begin{bmatrix} r & t' \\ t & r' \end{bmatrix}, \quad \begin{pmatrix} b_{13} \\ b_{24} \end{pmatrix} = \begin{bmatrix} r & t' \\ t & r' \end{bmatrix} \begin{pmatrix} a_{13} \\ a_{24} \end{pmatrix} \tag{3.2.4}$$

where the sub-matrices are given by

$$\begin{aligned} r &= r^{(1)} + t'^{(1)}[I - r^{(2)}r'^{(1)}]^{-1}r^{(2)}t^{(1)} \\ t' &= t'^{(1)}[I - r^{(2)}r'^{(1)}]^{-1}t'^{(2)} \\ t &= t^{(2)}[I - r'^{(1)}r^{(2)}]^{-1}t^{(1)} \\ r' &= r'^{(2)} + t^{(2)}[I - r'^{(1)}r^{(2)}]^{-1}r'^{(1)}t'^{(2)} \end{aligned} \tag{3.2.5}$$

3.2.1 Feynman paths

We can get insight into this result by expanding the expressions in a geometric series as follows:

$$\begin{aligned} t &= t^{(2)}\left[I - r'^{(1)}r^{(2)}\right]^{-1} t^{(1)} \\ &= t^{(2)}\left[I + r'^{(1)}r^{(2)} + (r'^{(1)}r^{(2)})^2 + \cdots\right] t^{(1)} \\ &= t^{(2)}t^{(1)} + t^{(2)}r'^{(1)}r^{(2)}t^{(1)} + t^{(2)}r'^{(1)}r^{(2)}r'^{(1)}r^{(2)}t^{(1)} + \cdots \end{aligned} \tag{3.2.6}$$

The first term is the amplitude for transmission through the two obstacles without any reflection, the second term for transmission with two reflections, the third term for transmission with four reflections and so on. Consider the element (m, n) of these terms

$$\left(t^{(2)}[r'^{(1)}r^{(2)}t^{(1)}]\right)_{mn} \to \sum_{m1}\sum_{m2}\sum_{m3}\left(t^{(2)}\right)_{m,m3}\left(r'^{(1)}\right)_{m3,m2}\left(r^{(2)}\right)_{m2,m1}\left(t^{(1)}\right)_{m1,n} \tag{3.2.7}$$

We could depict each term in this summation by a Feynman path. A path starts out in mode n, transmits to mode m_1, reflects to mode m_2, reflects again to mode m_3, and then transmit to mode m. The overall transmission amplitude from mode n to mode m can be expressed as

$$(t)_{mn} = \sum_{P\in\text{all paths}} A_P(\text{the amplitude for Path } P) \tag{3.2.8}$$

3.2.2 Combining successive sections incoherent

The transimission probability is obtained by squaring the transimission amplitude:

$$T_{mn} = t^*_{mn}t_{mn} = \sum_P A^*_P A_P + \sum_{P\neq P'}\sum_{P'} A^*_P A_P \tag{3.2.9}$$

The first term represents the sum of the probabilities of all the paths, while the second term is due to interference among the different paths. If the length of a section is much greater than the phase-relaxation length then sucessive sections could be treated as incoherent. This means theat instead of adding the amplitudes of the different paths, we should add their probabilities, which means dropping the cross-terms. The simplest way to do this is to calculate the transmission probability by combining probability matrices instead of S-matrices. Instead of $s = s_1 \otimes s_2$, we calculate $S = S_1 \otimes S_2$ where the probability matrix S is obtained simply by squaring the corresponding elements of the amplitude matrix s:

$$S(m, n) = |s(m, n)|^2 \tag{3.2.10}$$

One can gain insight into the effect of quantum interference by comparing the results obtained by combining successive scatters coherently and incoherently. A simple example involving two scatterers in a single-mode conductor whoese S-matrices are given by

$$s_1 = \begin{bmatrix} r_1 & t'_1 \\ t_1 & r'_1 \end{bmatrix}, \quad s_2 = \begin{bmatrix} r_2 & t'_2 \\ t_2 & r'_2 \end{bmatrix} \tag{3.2.11}$$

Since we are considering just one mode, the quantities r, t etc. are just complex numbers but not matrices. The probability matrices are given by

$$S_1 = \begin{bmatrix} R_1 & T_1 \\ T_1 & R_1 \end{bmatrix}, \quad S_2 = \begin{bmatrix} R_2 & T_2 \\ T_2 & R_2 \end{bmatrix} \tag{3.2.12}$$

where $R_i = |r_i|^2 = |r_i'|^2$ and $T_i = |t_i|^2 = |t_i'|^2$ for $i = 1, 2$. The unitarity of the S-matrix requires that $R_i + T_i = 1$. Combine the two individual amplitude matrices to obtain the composite transimission amplitude3.2.5

$$t = \frac{t_1 t_2}{1 - r_1' r_2} \tag{3.2.13}$$

Squaring this, we obtain the composite transmission probability

$$T = |t|^2 = \frac{T_1 T_2}{1 - 2\sqrt{R_1 R_2}\cos\theta + R_1 R_2} \quad \text{(coherent)} \tag{3.2.14}$$

where θ is the phase shift acquired in one round-trip between the scatterers: $\theta = \text{phase}(r_1') + \text{phase}(r_2')$. If we treat the two scatterers incoherently by combining the probability matrices, we obtain

$$T = \frac{T_1 T_2}{1 - R_1 R_2} \quad \text{(incoherent)} \tag{3.2.15}$$

3.3 Green's functions:a brief introduction

The S-matrix tells us the response at one lead due to an excitation at another. The Green's function is a more powerful concept that gives us excitation at any point due to an excitation at any other. For non-interacting transport, the only excitations we need to worry about are those due to waves incident from the leads. For such excitations, the Green's function and the S-matrix are related concepts and what we use is largely a matter of taste. Whenever the response R is related to the excitation S by a differential operator D_{op}

$$D_{op} R = S \tag{3.3.1}$$

We can define a Green's fuction and express the response in the forms

$$R = D_{op}^{-1} S = GS \quad \text{where} \quad G = D_{op}^{-1} \tag{3.3.2}$$

Out problem can be expressed in teh form

$$[E - H_{op}]\,\Psi = S \tag{3.3.3}$$

where Ψ is the wavefunction, and S is an equivalent excitation term due to a wave incident from one of the leads. The corresponding Green's function can be written as

$$G = [E - H_{op}]^{-1} \tag{3.3.4}$$

where H_{op} is the Hamiltonian operator, the subband energy E_s has been included as part of the potential $U(x, y)$

$$H_{op} = \frac{(\mathrm{i}\hbar\nabla + e\boldsymbol{A})^2}{2m} + U(\boldsymbol{r}) \tag{3.3.5}$$

3.3.1 Retarded and advanced Green's functions

The inverse of a different operator is not uniquely specified till we specify the boundary conditions. It is common to define two different Green's functions (retarded and advanced) corresponding to two different boundary conditions. The difference is best appreciated with a simple example. Consider a simple 1D wire with a constant potential energy U_0 and zero vector potential, thus we can write

$$G = \left[E - U_0 + \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\right]^{-1} \tag{3.3.6}$$

The Green's function satisfies the equation

$$\left[E - U_0 + \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\right] G(x,x') = \delta(x-x') \tag{3.3.7}$$

This looks like the schrodinger equation except for the source term $\delta(x-x')$ on the right. We could view the Green's function $G(x,x')$ as the wavefunction at x resulting from a unit rise to two waves traveling outwards from the point of excitation, with amplitudes A^+ and A^-. We can write

$$G(x,x') = \begin{cases} A^+ e^{ik(x-x')} & x > x' \\ A^- e^{-ik(x-x')} & x < x' \end{cases} \tag{3.3.8}$$

where $k = \sqrt{2m(E-U_0)}/\hbar$. Regardless of what A^+ and A^- might be, the Green's function must be continuous at $x = x'$ to satisfies the above equation. Thus we have

$$[G(x,x')]_{x=x'+} = [G(x,x')]_{x=x'-} \tag{3.3.9}$$

while the derivative must be discontinuous by $2m/\hbar^2$ to satisfy the delta function source term:

$$\left[\frac{\partial G(x,x')}{\partial x}\right]_{x=x'+} - \left[\frac{\partial G(x,x')}{\partial x}\right]_{x=x'-} = \frac{2m}{\hbar^2} \tag{3.3.10}$$

These two conditions allow us to solve for A^+ and A^-:

$$A^+ = A^-, \quad \mathrm{i}k\left[A^+ + A^-\right] = \frac{2m}{\hbar^2} \Rightarrow A^+ = A^- = \frac{m}{\mathrm{i}\hbar^2 k} \tag{3.3.11}$$

and the Green's function is given by

$$G^R(x,x') = -\frac{\mathrm{i}}{\hbar v}\exp\left(\mathrm{i}k|x-x'|\right) \tag{3.3.12}$$

It is important to note that there is another solution

$$G^A(x,x') = \frac{\mathrm{i}}{\hbar v}\exp\left(-\mathrm{i}k|x-x'|\right) \tag{3.3.13}$$

where

$$v = \frac{\hbar k}{m}, \quad k = \sqrt{\frac{2m(E-U_0)}{\hbar^2}} \tag{3.3.14}$$

which satisfies the same equation. This solution consists of incoming waves that disapper at the point of excitation rather than outgoing waves that originate at the point of excitation. The first solution is called the retarded Green's function G^R while the second is called the advanced Green's function G^A.

3.3.2 The infinitesimal

One way to incorporate the boundary conditions into the equation itself is to add an infinitesimal imaginary part to the energy

$$\left[E - U_0 + \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} \pm \mathrm{i}\eta\right] G^{R/A}(x,x') = \delta(x-x') \tag{3.3.15}$$

The small imaginary part ensures that the wavevector k has a small positive or negative imaginary part, which ensures that the waves are outgoing or incoming respectively.

$$k' = \sqrt{\frac{2m(E-U_0 \pm \mathrm{i}\eta)}{\hbar^2}} = \frac{\sqrt{2m(E-U_0)}}{\hbar}\sqrt{1 + \frac{\pm\mathrm{i}\eta}{E-U_0}} \approx k \pm \mathrm{i}\frac{m\eta}{\hbar^2 k} \equiv k(1 \pm \mathrm{i}\delta) \tag{3.3.16}$$

In general the retarded Green's function is defined As

$$G^R = [E - H_{op} + \mathrm{i}\eta]^{-1}, \quad \eta \to 0^+ \tag{3.3.17}$$

$$G^A = [E - H_{op} - \mathrm{i}\eta]^{-1}, \quad \eta \to 0^+ \tag{3.3.18}$$

3.3.3 Green's function for a multi-moded wire

Next let us look at the Green's function for a infinite multi-moded wire. The Green's function represents the wavefunction at (x, y) due to an excitation at (x', y'). We could write the Green's function in terms of the transverse modes $\chi_n(y)$ of the wire as

$$G^R(x, y; x', y') = \sum_m A_m^{\pm}\chi_m(y) \exp\left[\mathrm{i}k_m|x - x'|\right] \tag{3.3.19}$$

Where the A_m^+ and A_m^- are the amplitudes of the different modes that propagate away from the source. The transverse mode wavefunctions, $\chi_m(y)$, satisfy the equation (with $\boldsymbol{B} = 0$)

$$\left[\frac{-\hbar^2}{2m}\frac{\partial^2}{\partial y^2} + U(y)\right]\chi_m(y) = \epsilon_{m,0}\chi_m(y) \tag{3.3.20}$$

where $U(y)$ is the transverse confining potential in the y-direction. These functions are orthonormal:

$$\int \chi_m(y)\chi_n(y)\mathrm{d}y = \delta_{mn} \tag{3.3.21}$$

Since they satisfy the same equation with differnet eigenvalues. We will assume that these functions are real. To calculate the mode amplitudes $A_m^{\pm}$, we processed as we did for the single-mode wire.

$$\left[G^R(x, x')\right]_{x=x'+} = \left[G^R(x, x')\right]_{x=x'-} \tag{3.3.22}$$

$$\left[\frac{\partial G^R(x, x')}{\partial x}\right]_{x=x'+} - \left[\frac{\partial G^R(x, x')}{\partial x}\right]_{x=x'-} = \frac{2m}{\hbar^2}\delta(y - y') \tag{3.3.23}$$

we obtain

$$\begin{aligned}\sum_m A_m^+\chi_m(y) &= \sum_m A_m^-\chi_m(y)\\ \sum_m \mathrm{i}k_m\left(A_m^+ + A_m^-\right)\chi_m(y) &= \frac{2m}{\hbar^2}\delta(y - y')\end{aligned} \tag{3.3.24}$$

Multiplying both sides of these equations by $\chi_n(y)$ and integrating over y, we obtain

$$A_m^+ = A_m^-, \quad A_m^+ = A_m^- = \frac{m}{\mathrm{i}\hbar^2 k_m}\chi_m(y') = -\frac{\mathrm{i}}{\hbar v_m}\chi_m(y') \tag{3.3.25}$$

where $v_m = \hbar k_m/m$ is the velocity in mode m. Thus the retarded Green's function for a multi-moded wire is given by

$$G^R(x, y; x', y') = \sum_m -\frac{\mathrm{i}}{\hbar v_m}\chi_m(y)\chi_m(y') \exp\left(\mathrm{i}k_m|x - x'|\right) \tag{3.3.26}$$

where

$$k_m = \sqrt{\frac{2m(E - \epsilon_{m,0})}{\hbar^2}}, \quad v_m = \frac{\hbar k_m}{m} \tag{3.3.27}$$

3.3.4 Eigenfuction expansion

We end this section by deriving a result that is often used to calculate Green's functions. The basic idea is that for any structure, if we know the eigenfucntions of the Hamiltonian operator

$$H_{op}\psi_\alpha(\boldsymbol{r}) = \epsilon_\alpha\psi_\alpha(\boldsymbol{r}) \tag{3.3.28}$$

Then we can calculate the Green's function by performing the following summation:

$$G^R(\boldsymbol{r}, \boldsymbol{r}') = \sum_\alpha \frac{\psi_\alpha(\boldsymbol{r})\psi_\alpha^*(\boldsymbol{r}')}{E - \epsilon_\alpha + \mathrm{i}\eta}, \quad \eta \to 0^+ \tag{3.3.29}$$

The eigenfunctions satisfies the orthonormality condition

$$\int \psi_\alpha^*(\boldsymbol{r})\psi_\beta(\boldsymbol{r})\mathrm{d}\boldsymbol{r} = \delta_{\alpha\beta} \tag{3.3.30}$$

So that we can expand the Green's function in the form

$$G^R(\boldsymbol{r},\boldsymbol{r}') = \sum_\alpha C_\alpha(\boldsymbol{r}')\psi_\alpha(\boldsymbol{r}) \tag{3.3.31}$$

Substituting this into the defining equation for the Green's function

$$[E - H_{op} + \mathrm{i}\eta]\, G^R(\boldsymbol{r},\boldsymbol{r}') = \delta(\boldsymbol{r}-\boldsymbol{r}') \tag{3.3.32}$$

we obtain

$$\sum_\alpha C_\alpha(\boldsymbol{r}')\,[E - \epsilon_\alpha + \mathrm{i}\eta]\,\psi_\alpha(\boldsymbol{r}) = \delta(\boldsymbol{r}-\boldsymbol{r}') \tag{3.3.33}$$

Multiplying both sides by $\psi_\beta^*(\boldsymbol{r})$ and integrating over $\boldsymbol{r}$, we obtain

$$C_\beta(\boldsymbol{r}') = \frac{\psi_\beta^*(\boldsymbol{r}')}{E - \epsilon_\beta + \mathrm{i}\eta} \tag{3.3.34}$$

For advanced Green's function, we have

$$G^A(\boldsymbol{r},\boldsymbol{r}') = \sum_\alpha \frac{\psi_\alpha(\boldsymbol{r})\psi_\alpha^*(\boldsymbol{r}')}{E - \epsilon_\alpha - \mathrm{i}\eta}, \quad \eta \to 0^+ \tag{3.3.35}$$

It is straightforward to show that the advanced Green's function is the Hermitian conjugate of the retarded Green's function:

$$G^A(\boldsymbol{r},\boldsymbol{r}') = \left[G^R(\boldsymbol{r}',\boldsymbol{r})\right]^* \to G^A = \left[G^R\right]^\dagger \tag{3.3.36}$$

3.4 S-matrix and the Green's function

The Fisher-lee relation provides a direct connection between the S-matrix and the Green's funcion. Consider a conductor connected to a set of leads. For convenience, we use a different coordinate system in each lead as shown in Fig.3.4.1. The interface between lead p and the conductor is defined by $x_p = 0$. We will use the symbol G_{qp}^R to denote the Green's functioin between a point lying on the plane $x_p = 0$ and another point lying on $x_q = 0$:

$$G_{qp}^R(y_q; y_p) = G^R(x_q = 0, y_q; x_p = 0, y_p) \tag{3.4.1}$$

Neglect the transverse dimension (y) of the leads and treat them as 1D. We know that the unit excitationn at $x_p = 0$ gives rise to a wave of amplitude A_p^- away from the conductor and a wave of amplitude A_p^+ toward the conductor. The wave traveling toward the conductor is scattered by the conductor into different leads. Hence we can write

$$G_{qp}^R = \delta_{qp}A_p^- + s'_{qp}A_p^+ \tag{3.4.2}$$

But we know that

$$A_p^- = A_p^+ = -\frac{\mathrm{i}}{\hbar v_p} \tag{3.4.3}$$

also

$$s'_{qp} = s_{qp}\sqrt{\frac{v_p}{v_q}} \tag{3.4.4}$$

Hence we obtain

$$s_{qp} = -\delta_{qp} + \mathrm{i}\hbar\sqrt{v_q v_p}G_{qp}^R \tag{3.4.5}$$

3.4.1 Multi-moded leads

Take multiple modes into account, we can write the Green's function between lead p and lead q as

$$G^R_{qp}(y_q; y_p) = \sum_{m\in p}\sum_{n\in q}\left[\delta_{nm}A^-_m + s'_{nm}A^+_m\right]\chi_n(y_q) \tag{3.4.6}$$

we know that

$$A^-_m = A^+_m = -\frac{\mathrm{i}}{\hbar v_m} \tag{3.4.7}$$

also

$$s'_{nm} = s_{nm}\sqrt{\frac{v_m}{v_n}} \tag{3.4.8}$$

Hence we obtain

$$\begin{aligned} G^R_{qp}(y_q; y_p) &= \sum_{m\in p}\sum_{n\in q}\left[-\delta_{nm}\frac{\mathrm{i}}{\hbar v_m} + s_{nm}\frac{-\mathrm{i}}{\hbar}\sqrt{\frac{1}{v_n v_m}}\right]\chi_n(y_q)\chi_m(y_p) \\ &= \sum_{m\in p}\sum_{n\in q} -\frac{\mathrm{i}}{\hbar\sqrt{v_n v_m}}\chi_n(y_q)\left[\delta_{nm} + s_{nm}\right]\chi_m(y_p) \end{aligned} \tag{3.4.9}$$

In order to obtain an expression for an individual S-matrix element, we multiply equation above by $\chi_m(y_q)\chi_n(y_p)$ and integrate over y_q and y_p to obtain Fisher

$$s_{nm} = -\delta_{nm} + \mathrm{i}\hbar\sqrt{v_n v_m}\int \mathrm{d}y_q \int \mathrm{d}y_p \chi_n(y_q) G^R_{qp}(y_q; y_p)\chi_m(y_p) \tag{3.4.10}$$

3.5 Tight-binding model(or the method of finite differences)

Basically, we need too solve the differnetial equation for the Green's function

$$\left[E - H_{op} + \mathrm{i}\eta\right] G^R(\boldsymbol{r}, \boldsymbol{r}') = \delta(\boldsymbol{r} - \boldsymbol{r}') \tag{3.5.1}$$

for arbitrary $U(\boldsymbol{r})$ and $\boldsymbol{A}(\boldsymbol{r})$. We will restrict the discussion to 2D systems, but the method can be applied to 3D systems in a straightforward manner. A common approach for solving a differential equation is to discretize the spatial coordinate so that the Green's function becomes a matrix

$$G^R(\boldsymbol{r}_i, \boldsymbol{r}_j) \to G^R(i, j) \tag{3.5.2}$$

where the indices i, j denote points on a discrete lattice. The differential equation becomes a matrix equation

$$[(E + \mathrm{i}\eta)I - H]G^R = I \tag{3.5.3}$$

where H is the matrix representation of the Hamiltonian operator H_{op}. I is the identity matrix.

3.5.1 Matrix representation for the Hamiltonian operator in 1D

In 1D, with vector potential set to zero, the Hamiltonian operator simplifies to

$$H_{op} = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{x^2} + U(x) \tag{3.5.4}$$

To obtain a matrix representation for this operator, we consider the quantity $H_{op}F(x)$ where $F(x)$ is any function of x. Now we choose a discrete lattice whose points are located at $x = ja$, j being an integer and write

$$[H_{op}F(x)]_{x=ja} = -\frac{\hbar^2}{2m}\left[\frac{\mathrm{d}^2F(x)}{\mathrm{d}x^2}\right]_{x=ja} + U_jF_j \tag{3.5.5}$$

where

$$U_j = U(ja), \quad F_j = F(ja) \tag{3.5.6}$$

Assuming a is small, we can approximate the second derivative by

$$\left[\frac{\mathrm{d}^2F(x)}{\mathrm{d}x^2}\right]_{x=ja} \approx \frac{F_{j+1} - 2F_j + F_{j-1}}{a^2} \tag{3.5.7}$$

With this approximation we can write

$$\begin{aligned}
[H_{op}F(x)]_{x=ja} &= -\frac{\hbar^2}{2ma^2}(F_{j+1} - 2F_j + F_{j-1}) + U_jF_j \\
&= -t(F_{j+1} + F_{j-1}) + (2t + U_j)F_j, \quad t = \frac{\hbar^2}{2ma^2} \\
&= \sum_i H(j,i)F_i
\end{aligned} \tag{3.5.8}$$

where

$$H(j,i) = \begin{cases} -t & i = j \pm 1 \\ 2t + U_j & i = j \\ 0 & \text{otherwise} \end{cases} \tag{3.5.9}$$

The desired matrix representation for the Hamiltonian operator in 1D is thus obtained.

$$H = \begin{bmatrix} 2t + U_1 & -t & 0 & 0 & \cdots \\ -t & 2t + U_2 & -t & 0 & \cdots \\ 0 & -t & 2t + U_3 & -t & \cdots \\ 0 & 0 & -t & 2t + U_4 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} \tag{3.5.10}$$

3.5.2 Dispersioin relation fro a discrete lattice

We know that for a uniform 1D wire with a constant potential $U(x) = U_0$, the eigenfunctions are plane waves with a parabolic dispersion relation:

$$\psi_k(x) = \exp(\mathrm{i}kx), \quad E(k) = U_0 + \frac{\hbar^2k^2}{2m} \tag{3.5.11}$$

For the discrete lattice model, we can write the Schrodinger equation as

$$E\psi_j = (U_0 + 2t)\psi_j - t\psi_{j+1} - t\psi_{j-1}) \tag{3.5.12}$$

which is satisfied by a solution of the form

$$\psi_j = \exp(\mathrm{i}kx_j), \quad x_j = ja \tag{3.5.13}$$

privided

$$E(k) = U_0 + 2t[1 - \cos(ka)] = U_0 + \frac{\hbar^2}{ma^2}\sin^2\left(\frac{ka}{2}\right) \tag{3.5.14}$$

As a tends to zero, the dispersion relation approaches the parabolic form. The velocity is given by

$$v(k) = \frac{1}{\hbar}\frac{\partial E}{\partial k} = \frac{2at}{\hbar}\sin(ka) \tag{3.5.15}$$

As a tends to zero, this expression approaches the familiar form $v = \hbar k/m$.

3.5.3 Matrix representation for the Hamiltonian operator in 2D

In 2D, the matrix elements of Hamiltonian matrix simplifies to

$$H_{ij} = \begin{cases} -\tilde{t}_{ij} & \text{if } i \text{ and } j \text{ are nearest neighbors} \\ zt + U_i & i = j \\ 0 & \text{otherwise} \end{cases} \tag{3.5.16}$$

where z is the coordination number (number of nearest neighbors) of the lattice. For a square lattice, $z = 4$. If the vector potential is zero, then the nearest neighbor coupling is equal to $-t$ as in the 1D example. With a non-zero vector potential it is modified to

$$\tilde{t}_{ij} = t \exp\left(\frac{\mathrm{i}e}{\hbar}\int_{r_i}^{r_j} \boldsymbol{A}(\boldsymbol{r}) \cdot \mathrm{d}\boldsymbol{r}\right) \approx t \exp\left(\frac{\mathrm{i}e}{\hbar}\boldsymbol{A}\left(\frac{\boldsymbol{r}_i + \boldsymbol{r}_j}{2}\right) \cdot (\boldsymbol{r}_j - \boldsymbol{r}_i)\right) \tag{3.5.17}$$

3.5.4 Truncating the matrix

Now that we have a matrix representation for the Hamiltonian operator it may seem straightforward to set up the matrix $[(E + \mathrm{i}\eta)I - H]$ and invert it.

$$G^R = [(E + \mathrm{i}\eta)I - H]^{-1} \tag{3.5.18}$$

If we simply truncate the matrix to a finite size, we would effectively be describing a closed system with fully relfecting boundaries and not the open system with non-reflecting boundaries that we would like to describe. Consider a conductor connected to the lead p as shown in Fig.3.5.2. We can partition the overall Green's function into submatrices as follows:

$$\begin{bmatrix} G_p & G_{pC} \\ G_{Cp} & G_C \end{bmatrix} = \begin{bmatrix} (E + \mathrm{i}\eta)I - H_p & \tau_p \\ \tau_p^\dagger & EI - H_C \end{bmatrix}^{-1} \tag{3.5.19}$$

where the matrix $[(E + \mathrm{i}\eta)I - H_p]$ represents the isolated lead, while the matrix $[EI - H_C]$ represents the isolated conductor. We could add an infinitesimal imaginary term ($\mathrm{i}\eta$) as well but it is not necessary. The coupling matrix is non-zero only for adjecent points i and p_i

$$\tau_p(i, p_i) = t \tag{3.5.20}$$

The submatrix G_C represents the Green's function for the conductor taking into account its coupling to the lead p. From the definition of the inverse of a matrix, we have

$$[(E + \mathrm{i}\eta)I - H_p]G_{pC} + [\tau_p]G_C = 0 \tag{3.5.21}$$

$$[EI - H_C]G_C + [\tau_p^\dagger]G_{pC} = I \tag{3.5.22}$$

From the first equation, we obtain

$$G_{pC} = -g_p^R \tau_p G_C, \quad g_p^R = [(E + \mathrm{i}\eta)I - H_p]^{-1} \tag{3.5.23}$$

g_p^R is the Green's function for the isolated semi-infinite lead p. Substituting this into the second equation, we obtain

$$G_C = \left[EI - H_C - \tau_p^\dagger g_p^R \tau_p\right]^{-1} \tag{3.5.24}$$

Note that the matrices in this expression are finite matrices of dimensions $(C \times C)$, C being the number of points inside the conductor. The effect of the semi-infinite lead is included through the self-energy term

$$\Sigma_p^R = \tau_p^\dagger g_p^R \tau_p \tag{3.5.25}$$

with zero vector potential, $\tau_p = -t$, we have

$$G^R = G_C = [EI - H_C - \Sigma^R]^{-1}, \quad \Sigma^R = \sum_p \Sigma_p^R, \quad \Sigma_p^R(i, j) = t^2 g_p^R(p_i, p_j) \tag{3.5.26}$$

3.5.5 Self-energy

For a semi-infinite discrete wire it can be shwon that the Green's function between two points along the edge is given by

$$g_p^R(p_i, p_j) = -\frac{1}{t}\sum_m \chi_m(p_i)\exp(\mathrm{i}k_m a)\chi_m(p_j) \tag{3.5.27}$$

where the (real) function $\chi_m(p_i)$ describes the transverse profile of mode m in lead p and v_m is its velocity. Substituting this into the expression for the self-energy, we obtain

$$\Sigma_p^R(i, j) = -t\sum_m \chi_m(p_i)\exp(\mathrm{i}k_m a)\chi_m(p_j) \tag{3.5.28}$$

3.5.6 Transmission function

Use the Fisher-lee relation, we can write the transmission function between lead p and lead q as

$$\overline{T}_{qp} = \mathrm{tr}\left[\Gamma_q G^R \Gamma_p G^A\right] \tag{3.5.29}$$

where

$$\Gamma_p = \sum_m \chi_m(p_i)\frac{\hbar v_m}{a}\chi_m(p_j) = \mathrm{i}\left[\Sigma_p^R - \Sigma_p^A\right], \quad \Sigma_p^A = \left[\Sigma_p^R\right]^\dagger \tag{3.5.30}$$

Proof. From the Fisher-lee relation, we have

$$s_{nm} = -\delta_{nm} + \mathrm{i}\hbar\sqrt{v_n v_m}\int \mathrm{d}y_q \int \mathrm{d}y_p \chi_n(y_q) G_{qp}^R(y_q; y_p)\chi_m(y_p) \tag{3.5.31}$$

We can write for $n \neq m$

$$|s_{nm}|^2 = \frac{\hbar^2 v_n v_m}{a^2}\sum_{i,j,i',j'} \chi_n(q_j) G^R(j,i)\chi_m(p_i)\chi_n(q_{j'}) G^A(i',j')\chi_m(p_{i'}) \tag{3.5.32}$$

where we have made use of the relation $\int \mathrm{d}y_q \to a\sum_j$ and $\int \mathrm{d}y_p \to a\sum_i$ and $G^A = [G^R]^*$. Summing over all modes in lead p and lead q, we obtain

$$\begin{aligned}\overline{T}_{qp} &= \sum_{m\in p}\sum_{n\in q}|s_{nm}|^2 \\ &= \frac{\hbar^2}{a^2}\sum_{i,j,i',j'} G^R(j,i)\left[\sum_{n\in q} v_n\chi_n(q_j)\chi_n(q_{j'})\right] G^A(i',j')\left[\sum_{m\in p} v_m\chi_m(p_i)\chi_m(p_{i'})\right] \\ &= \mathrm{tr}\left[\Gamma_q G^R \Gamma_p G^A\right]\end{aligned} \tag{3.5.33}$$

where

$$\Gamma_p(i,i') = \sum_{m\in p}\chi_m(p_i)\frac{\hbar v_m}{a}\chi_m(p_{i'}), \quad \Gamma_q(j,j') = \sum_{n\in q}\chi_n(q_j)\frac{\hbar v_n}{a}\chi_n(q_{j'}) \tag{3.5.34}$$

□

The Green's function G^R describes the dynamics of the electrons inside the conductor, taking the effect of the leads into account through the self-energy terms. The function Γ describes the coupling between the conductor and the leads.

3.5.7 Sum rule

Earlier in this chapter, we derived a sum rule for the transimission function

$$\sum_q \overline{T}_{pq} = \sum_q \overline{T}_{qp} = M_p \tag{3.5.35}$$

where M_p is the number of modes in lead p. This sum rule ensures current conservation. We can derive a similar sum rule in terms of the spectral function.

$$\begin{aligned} \sum_q \overline{T}_{pq} &= \mathrm{tr}[\Gamma_p G^R (\sum_q \Gamma_q) G^A] = \mathrm{tr}[\Gamma_p G^R \Gamma G^A] \\ \sum_q \overline{T}_{qp} &= \mathrm{tr}[\sum_q \Gamma_q G^R \Gamma_p G^A] = \mathrm{tr}[\Gamma G^R \Gamma_p G^A] \end{aligned} \tag{3.5.36}$$

where

$$\Gamma = \sum_q \Gamma_q = \sum_q \mathrm{i}[\Sigma_q^R - \Sigma_q^A] = \mathrm{i}[\Sigma^R - \Sigma^A] \tag{3.5.37}$$

we obtain the desired sum rule

$$\sum_q \overline{T}_{pq} = \sum_q \overline{T}_{qp} = \mathrm{tr}[\Gamma_p A] = M_p \tag{3.5.38}$$

if we make use of the following identity:

$$A \equiv \mathrm{i}[G^R - G^A] = G^R \Gamma G^A = G^A \Gamma G^R \tag{3.5.39}$$

The quantity A is called the spectral function. It describes the density of states inside the conductor taking into account the coupling to the leads.

3.6 Self-energy

We have seen in the last section that the Green's function which is defined as

$$G^R = [E - H + \mathrm{i}\eta]^{-1} \tag{3.6.1}$$

can be modified to take into account the effect of the leads by introducing a self-energy term Σ^R:

$$G^R = \left[E - H_C - \Sigma^R\right]^{-1} \tag{3.6.2}$$

Where H_C is the Hamiltonian matrix for the isolated conductor, and the self-energy σ^R describes the effect of the leads on the conductor. This is an important conceptual step, for it allows us to replace an infinite open system with a finite one. There are two important concepts defined by the relationship

$$\Gamma \equiv \mathrm{i}\left[\Sigma^R - \Sigma^A\right] \tag{3.6.3}$$

$$A \equiv \mathrm{i}\left[G^R - G^A\right] \tag{3.6.4}$$

and derive a useful relationship between them:

$$A = G^R \Gamma G^A = G^A \Gamma G^R \tag{3.6.5}$$

which is the analog of the unitarity relation

$$I = SS^\dagger = S^\dagger S \tag{3.6.6}$$

3.6.1 Eigenstate lifetime

In isolated or closed systems whose eigenstates are found by diagonalizing the Hamiltonian H_C:

$$H_C\psi_{\alpha,0} = \epsilon_{\alpha,0}\psi_{\alpha,0} \tag{3.6.7}$$

Taking the coupling to the leads into account through the self-energy term, the eigenstates are found by diagonalizing the effective Hamiltonian

$$H_{eff} = H_C + \Sigma^R, \quad \left[H_C + \Sigma^R\right]\psi_\alpha = \epsilon_\alpha\psi_\alpha \tag{3.6.8}$$

There is an important difference between these two sets of eigenstates. The eigenvalues $\epsilon_{\alpha,0}$ of the isolated conductor are real, while the eigenvalues ϵ_α of the open conductor are complex, since the self-energy is in general not Hermitian. We could write the new eigenenergies as

$$\epsilon_\alpha = \epsilon_{\alpha,0} - \Delta_\alpha - \mathrm{i}(\gamma_\alpha/2) \tag{3.6.9}$$

where $\epsilon_{\alpha,0}$ is the eneegy of the isolated conductor described by H_C. The time dependence of an eigenstate of H_{eff} is given by

$$\psi_\alpha(t) = \psi_\alpha(0)\exp\left(-\frac{\mathrm{i}\epsilon_\alpha t}{\hbar}\right) = \psi_\alpha(0)\exp\left(-\frac{\mathrm{i}(\epsilon_{\alpha,0}-\Delta_\alpha)t}{\hbar}\right)\exp\left(-\frac{\gamma_\alpha t}{2\hbar}\right) \tag{3.6.10}$$

Δ_α represents a shift in the energy of the eigenstate due to the coupling to the leads, while γ_α represents a finite lifetime of the eigenstate due to escape into the leads.

$$|\psi_\alpha(t)|^2 = |\psi_\alpha(0)|^2\exp\left(-\frac{\gamma_\alpha t}{\hbar}\right) \tag{3.6.11}$$

The lifetime is given by $\tau_\alpha = \hbar/\gamma_\alpha$, which is infinite if the self-energy is Hermitian, that is, if $\Gamma = 0$.

3.6.2 Eigenfunction expansion

We can use the eigenfunctions of the effective Hamiltonian H_{eff} to expand the Green's function as follows:

$$G^R(\boldsymbol{r},\boldsymbol{r}') = \sum_\alpha \frac{\psi_\alpha(\boldsymbol{r})\psi_\alpha^\dagger(\boldsymbol{r}')}{E-\epsilon_\alpha} \tag{3.6.12}$$

if we define ψ_α as the eigenstate of H_{eff}.

$$\left[H_C + \Sigma^R\right]\psi_\alpha = \epsilon_\alpha\psi_\alpha \tag{3.6.13}$$

It is not correct for the effective Hamiltonian is not Hermitian. Hence the eigenstates are not orthogonal in the usual sense. Instead, we need to introduce the concept of bi-orthogonality. We define the left eigenstates ϕ_α as the eigenstates of the Hermitian conjugate of the effective Hamiltonian:

$$\left[H_C + \Sigma^A\right]\phi_\alpha = \epsilon_\alpha^*\phi_\alpha \tag{3.6.14}$$

The left and right eigenstates satisfy the bi-orthogonality condition

$$\int \phi_\alpha(\boldsymbol{r})\psi_\beta^*(\boldsymbol{r})\mathrm{d}\boldsymbol{r} = \delta_{\alpha\beta} \tag{3.6.15}$$

With this definition, we can expand the Green's function as

$$G^R(\boldsymbol{r},\boldsymbol{r}') = \sum_\alpha \frac{\psi_\alpha(\boldsymbol{r})\phi_\alpha^*(\boldsymbol{r}')}{E-\epsilon_\alpha} \tag{3.6.16}$$

3.6.3 Spectral function

We can write the spectral function in terms of the eigenstates of the effective Hamiltonian as follows:

$$\begin{aligned} A(\boldsymbol{r},\boldsymbol{r}') &= \mathrm{i}\left[G^R(\boldsymbol{r},\boldsymbol{r}') - G^A(\boldsymbol{r},\boldsymbol{r}')\right] \\ &= \sum_\alpha \psi_\alpha(\boldsymbol{r})\phi^*_\alpha(\boldsymbol{r}')\left[\frac{\mathrm{i}}{E-\epsilon_\alpha} - \frac{\mathrm{i}}{E-\epsilon^*_\alpha}\right] \\ &= \sum_\alpha \psi_\alpha(\boldsymbol{r})\phi^*_\alpha(\boldsymbol{r}')\frac{\gamma_\alpha}{(E-\epsilon_{\alpha 0}+\Delta_\alpha)^2+(\gamma_\alpha/2)^2} \end{aligned} \tag{3.6.17}$$

The spectral function evolves as the coupling to the leads is increased is illustrated in Fig.3.6.2. The self-energy Σ^R becomes larger and the shifts the eigenenergies. As a result, the peaks of the quantity $A(\boldsymbol{r},\boldsymbol{r})$ shift by Δ_α and broaden by an amount γ_α. When the coupling is very large, the peaks all merge and the peaks corresponding to different eigenstates can no longer be distinguished.
The trace of the spectral function represents the density of states:

$$N(E) = \frac{1}{2\pi}\,\mathrm{tr}\,A(E) \tag{3.6.18}$$

$$\mathrm{tr}\,A \equiv \int A(\boldsymbol{r},\boldsymbol{r})\mathrm{d}\boldsymbol{r} = \sum_\alpha \frac{\gamma_\alpha}{(E-\epsilon_{\alpha 0}+\Delta_\alpha)^2+(\gamma_\alpha/2)^2} \tag{3.6.19}$$

thus the density of states can be written as

$$\begin{aligned} N(E) &= \sum_\alpha \frac{1}{2\pi}\frac{\gamma_\alpha}{(E-\epsilon_{\alpha 0}+\Delta_\alpha)^2+(\gamma_\alpha/2)^2} \\ &\xrightarrow{\gamma_\alpha\to 0} \sum_\alpha \delta(E-\epsilon_{\alpha 0}+\Delta_\alpha) \end{aligned} \tag{3.6.20}$$

3.6.4 Local density of states

The diagonal elements of the spectral function give the local density of states

$$\begin{aligned} \rho(\boldsymbol{r},E) &= \frac{1}{2\pi}A(\boldsymbol{r},\boldsymbol{r},E) = -\frac{1}{\pi}\,\mathrm{Im}[G^R(\boldsymbol{r},\boldsymbol{r},E)] \\ &\sim \sum_\alpha \frac{1}{2\pi}\frac{\gamma_\alpha}{(E-\epsilon_{\alpha 0}+\Delta_\alpha)^2+(\gamma_\alpha/2)^2}\psi_\alpha(\boldsymbol{r})\phi^*_\alpha(\boldsymbol{r}) \\ &\to \sum_\alpha \delta(E-\epsilon_{\alpha 0})|\psi_\alpha(\boldsymbol{r})|^2, \quad \gamma_\alpha \to 0 \end{aligned} \tag{3.6.21}$$

3.6.5 A useful identity

There is a useful identity involving the spectral function that is often used in quantum transport calculations:

$$A = G^R\Gamma G^A = G^A\Gamma G^R \tag{3.6.22}$$

Proof.

$$[G^R]^{-1} - [G^A]^{-1} = \left[E - H_C - \Sigma^R\right] - \left[E - H_C - \Sigma^A\right] = \Sigma^A - \Sigma^R = \mathrm{i}\Gamma \tag{3.6.23}$$

Multiplying by G^R from the left and by G^A from the right, we obtain

$$G^A - G^R = \mathrm{i}G^R\Gamma G^A \to A = G^R\Gamma G^A \tag{3.6.24}$$

Similarly, multiplying by G^A from the left and by G^R from the right, we obtain

$$G^A - G^R = \mathrm{i}G^A\Gamma G^R \to A = G^A\Gamma G^R \tag{3.6.25}$$

□

3.7 Relation to other formalism

The transimission function $\overline{T}_{pq}$ from lead q to lead p can be written in terms of the Green's function as

$$\overline{T}_{pq} = \text{tr}\left[\Gamma_p G^R \Gamma_q G^A\right] \tag{3.7.1}$$

The Green's function G^R describes the dynamics of the electrons inside the conductor (taking the leads into account), while the Γ_p describes the strength of the coupling of lead p to the conductor.

3.7.1 Kubo formalism

The Kubo formalism is based on the fluctuation-dissipation theorem which relates the equilibrium noise (that is, fluctuatios) to the linear response conductance (that is, dissipation). A special case of this general theorem is the Nyquist-Johnson formula

$$G = \frac{1}{2k_B T}\int_{-\infty}^{\infty} \langle I(t+t_0)I(t_0)\rangle_{eq} \mathrm{d}t \tag{3.7.2}$$

This relationship allows us to calculate a non-equilibrium property like conductance indirectly by calculating the equilibrium noise using the methods of equilibrium statictical mechanics. The Kubo formula is commonly used to derive the following expression for the zero temperature conductivity of a uniform rectangular conductor with all sides equal to L (d is number of dimensions and all quantities are evaluated at the Fermi energy)

$$\sigma = \frac{2e^2}{h}\frac{\hbar^2}{L^d}\sum_{k,k'} v_x(k')v_x(k)\left|G^R(k',k)\right|^2 \tag{3.7.3}$$

is used as the starting point for calculating the ensemble-averaged conductivity of large conductors. We can write the conduntance as

$$G = \frac{\sigma L^{d-1}}{L} = \frac{2e^2}{h}\frac{\hbar^2}{L^2}\sum_{k,k'} v_x(k')v_x(k)\left|G^R(k',k)\right|^2 \tag{3.7.4}$$

which follows the Landauer formula $G = (2e^2/h)\overline{T}$ with the transmission function 3.5.29. Although 3.5.29 was derived using a discrete lattice in real space, we can use a unitary transformation to transform it to any other convenient representation. For example in the k-representation, we have

$$G = \frac{2e^2}{h}\sum_{k_1,k_2,k,k'} \Gamma_1(k_1,k')G^R(k',k)\Gamma_2(k,k_2)G^A(k_2,k_1) \tag{3.7.5}$$

Note that the terms in this summation with $k_1 = k'$ and $k_2 = k$

$$\Gamma_1(k',k')G^R(k',k)\Gamma_2(k,k)G^A(k,k') \tag{3.7.6}$$

are all positive real quantities since $G^A(k,k') = G^R(k',k)^*$ and the diagonal elements of Γ are real since Γ is Hermitian. By contrast the other terms with $k_1 \neq k$ and $k_2 \neq k'$ are complex quantities with randomly varying phases that average out to zero, and can be omitted from the summation. Hence we obtain

$$G = \frac{2e^2}{h}\sum_{k,k'} \Gamma_1(k',k')\left|G^R(k',k)\right|\Gamma_2(k,k) \tag{3.7.7}$$

This is identical to the result we are trying to prove if

$$\Gamma_1(k',k') = \Gamma_2(k,k) = \frac{\hbar v_x(\boldsymbol{k})}{L} \tag{3.7.8}$$

We will now justify this expression for Γ for a uniform conductor with two identical leads. The physical significance of the function $\Gamma_p(k,k)$ is that it represents $\hbar/2$ times the rate at which an electron placed in a state k will escape into lead p. From this point of view, the equation above is reasonable since the electron wavefunction is spread out over a box of length L and the electron can only escape through the two surfaces with a velocity v_x, the rate of escape is given by $2v_x/L$, so that

$$\Gamma_p(k,k) = \frac{\hbar}{2}\frac{2v_x(k)}{L} = \frac{\hbar v_x(k)}{L} \tag{3.7.9}$$

Consider a 1D conductor with N sites as shown in Fig.3.7.1. The only non-zero components of the self-energy involve the end-points. Assuming both leads to be identical we can write

$$\Gamma_1(1,1) = \frac{\hbar v}{2} = \Gamma_2(N,N) \tag{3.7.10}$$

Now we transform from the site representation to a k-representation using the unitary transformation

$$|k\rangle = \frac{1}{\sqrt{N}}\sum_j \exp(-\mathrm{i}kx_j)\,|j\rangle \tag{3.7.11}$$

Using the usual rules for matrix transformation

$$\Gamma_p(k,k) = \sum_{i,j}\langle k|i\rangle\,\Gamma_p(i,j)\,\langle j|k\rangle = \frac{1}{N}\sum_{i,j}\exp(-\mathrm{i}kx_i)\Gamma_p(i,j)\exp(\mathrm{i}kx_j) \tag{3.7.12}$$

we obtain (note that $L = Na$)

$$\Gamma_1(k,k) = \frac{1}{N}\Gamma_1(1,1) = \frac{\hbar v}{2N} = \frac{\hbar v}{L} = \Gamma_2(k,k) \tag{3.7.13}$$

Extend this result beyond 1D, we obtain the desired result

$$\Gamma_1(m,k;m,k) = \frac{\hbar v_m}{L} = \Gamma_2(m,k;m,k) \tag{3.7.14}$$

3.7.2 Transfer Hamiltonian formalism

The transfer Hamiltonian formalism is widely used to describe tunneling processes involving electron transfer between two leads separated by an insulator (Fig.3.7.2). In the transfer Hamiltonian formalism the current is related to the matrix elements, M, between lead 1 and lead 2

$$I = \frac{4\pi e}{\hbar}\int\Big[f_1(E) - f_2(E)|M(E)|^2\rho_1(E)\rho_2(E)\Big]\,\mathrm{d}E \tag{3.7.15}$$

where $\rho_1(E)$ and $\rho_2(E)$ are the density of states in lead 1 and lead 2 respectively. The change in the current in response to a change in the potential μ_2 (keeping μ_1 constant) can be written as

$$\frac{\partial I}{\partial\mu_2} = \frac{4\pi e}{\hbar}\Big[|M(E)|^2\rho_1(E)\rho_2(E)\Big]_{E=\mu_2} \tag{3.7.16}$$

where we have assumed low temperature (so that $f_2(E) \approx \theta(\mu_2 - E)$) and also neglected any change in M, ρ_1 and ρ_2 due to the applied bias. Relating the slope of the current-voltage curve to the density of states in the leads allows one to use current-voltage measurements to deduce the density of states $rho_2(E)$ in one lead if the density of states $\rho_1(E)$ in the other lead is known. In scattering formalism the current is given by

$$I = \frac{2e}{h}\int\overline{T}(E)\,[f_1(E) - f_2(E)]\,\mathrm{d}E \tag{3.7.17}$$

Thus we have

$$\overline{T}(E) = 4\pi^2|M(E)|^2\rho_1(E)\rho_2(E) \tag{3.7.18}$$

3.8 Feynman paths

In this section we show how the Green's function can be visualized in terms of Feynman paths. The Green's function is given by the inverse of the matrix

$$G^R = [E - H + \mathrm{i}\eta]^{-1} \tag{3.8.1}$$

The basic idea is to write the maatrix $[G^R]^{-1}$ as a sum of two matrices

$$[G^R]^{-1} = [G_0]^{-1} - \alpha \tag{3.8.2}$$

one of which represents the unperturbed system (G_0^{-1}) and the other represents the perturbation (α). We can write

$$G^R = \left[G_0^{-1} - \alpha\right]^{-1} = \left[G_0^{-1}(I - G_0\alpha)\right]^{-1} = (I - G_0\alpha)^{-1}G_0 \tag{3.8.3}$$

The inverse can be expanded in an infinite geometric series to obtain the following expression for G.

$$G^R = \left[I + G_0\alpha + (G_0\alpha)^2 + (G_0\alpha)^3 + \cdots\right] G_0 = G_0 + G_0\alpha G_0 + G_0\alpha G_0\alpha G_0 + \cdots \tag{3.8.4}$$

Consider the element (i, j) of one of the terms in this series:

$$G_0(i,i)\alpha(i,A)G_0(A,A)\alpha(A,j)G_0(j,j) \tag{3.8.5}$$

We could visualize this term as the amplitude of a 'path' going from j to a point A and then on to i. The total Green's function $G^R(i,j)$ is equal to the sum of the amplitudes of all such paths leading from j to i

$$G^R(i,j) = \sum_P A_P(i,j), \quad P \in \text{all paths from } j \text{ to } i \tag{3.8.6}$$

Each path is composed of series of segments which alternately involve unperturbed propagation (described by G_0) and a transition induced by the perturbation (described by α). The choice of G_0 and α is clearly not unique and has to be motivated by the physics of the problem at hand. For example, suppose we associate the unperturbed part with the diagonal elements of H_C

$$[G_0^{-1}]_{ij} = (E - U_i - 2t)\delta_{ij} \tag{3.8.7}$$

then G_0 represents the Green's function for a set of isolated lattice sites:

$$[G_0]_{ij} = \frac{\delta_{ij}}{E - U_i - 2t} \tag{3.8.8}$$

The perturbation consists of the hopping matrix element connecting adjacent sites:

$$\alpha_{ij} = t\exp[\mathrm{i}e\boldsymbol{A}\cdot(\boldsymbol{r}_i - \boldsymbol{r}_j)/\hbar] \quad \text{if } i \text{ and } j \text{ are nearest neighbors} \tag{3.8.9}$$

We are neglecting the effect of the leads which is represented by the self-energy term Σ^R. With this choice of G_0 and α, a Feynman path has the form shown in Fig.3.8.1.

3.8.1 Aharonov-Bohm (A-B) effect

Using a small ring 820nm in diameter etched out of a high quality gold film. The conductance of the ring was observed to oscillate as a function of the magnetic flux enclosed by the ring.

$$G = G_0 + \hat{G}\cos\left(\frac{|e|BS}{\hbar} + \overline{\phi}\right) (S \equiv \text{area enclosed by ring}) \tag{3.8.10}$$

The period B_p of the oscillations is obtained by the setting the phase difference to 2π

$$\frac{|e|BS}{\hbar} = 2\pi \Rightarrow B_pS = h/|e| \tag{3.8.11}$$

This is called the h/e effect since the period corresponds to a change in the enclosed flux by h/e. The transmission from mode m in lead 1 to mode n in lead 2 can be written as

$$T(n \leftarrow m) \rightarrow |t_1 + t_2|^2 \tag{3.8.12}$$

where t_1 is the total amplitude of all the Feynman paths going through the upper arm of the ring that start in mode m at A and end in mode n at B. Similarly t_2 is the total amplitude of all the paths going through the lower arm of the ring that start in mode m at A and end in mode n at B. The oscillations in the conductance arise because a vector potential changes the phase relationship between t_1 and t_2. The conductance peaks whenever t_1 and t_2 are in phase. The amplitude associated with a path connecting two nearest neighbors i and j is modified as follows

$$\alpha_{ij} = t\exp\left[\mathrm{i}e\boldsymbol{A}\cdot(\boldsymbol{r_i} - \boldsymbol{r_j})/\hbar\right] \tag{3.8.13}$$

Consequently the vector potential $\boldsymbol{A}$ introduces an extra phase to every path:

$$t_1(B) \sim t_1(0)\exp(\mathrm{i}\phi_1), \quad t_2(B) \sim t_2(0)\exp(\mathrm{i}\phi_2) \tag{3.8.14}$$

where

$$\phi_1 = \frac{e}{\hbar}\int_{\text{path 1}} \boldsymbol{A}\cdot \mathrm{d}\boldsymbol{l}, \quad \phi_2 = \frac{e}{\hbar}\int_{\text{path 2}} \boldsymbol{A}\cdot \mathrm{d}\boldsymbol{l} \tag{3.8.15}$$

We are assuming that the ring is thin enough that there is not much difference between the area enclosed by the inner and the outer circumference. The difference between the two phases is proportional to the flux enclosed by the ring:

$$\phi_1 - \phi_2 = \frac{|e|}{\hbar}\oint \boldsymbol{A}\cdot \mathrm{d}\boldsymbol{l} = \frac{|e|BS}{\hbar} \tag{3.8.16}$$

(B: magnetic field and S: area enclosed by ring). Hence

$$T(n \leftarrow m) = \left|t_1(0)e^{\mathrm{i}\phi_1} + t_2(0)e^{\mathrm{i}\phi_2}\right|^2 = |t_1(0)|^2 + |t_2(0)|^2 + 2|t_1(0)t_2(0)|\cos\left(\frac{|e|BS}{\hbar} + \phi\right) \tag{3.8.17}$$

Chapter 4

Quantum Hall effect

At high magnetic fields the Hall resistance has been observed to be quantized in units of h/e^2 with an accuracy that is specified in parts per million. This impressive accuracy arises from the near complete suppression of momentum relaxation processes in the quantum Hall regime, resulting in a truly ballistic conductor of incredibly high quality. Mean free paths of several millimeters have been observed, arising from that at high magnetic fields the electronic states carrying current in one direction are localized on one side of the sample while those carrying current in the other direction are localized on the other side of the sample.

4.1 Origin of 'zero' resistance

We know that at high magnetic fields the longitudinal resistance (measured using a macroscopic Hall bridge) oscillates as a function of the magnetic field. The density of states at high magnetic fields develops sharp peaks spaced by $\hbar\omega_c$ and the resistivity oscillates as the position of these peaks is changed relative to the Fermi energy. This can be done either by changing the magnetic field or by keeping the magnetic field fixed and changing the electron density (and hence the Fermi energy) by means of a gate voltage.
Intuitively it might appear that the resistance should be a minimum whenever the Fermi energy coincides with a peak in the density of states, that is, with a Landau level. However, the correct answer is just the opposite. The resistance is a minimum when the Fermi energy lies between two Landau levels so that the density of states at the Fermi energy is a minimum! But how does a smaple carry any current unless there are states at the Fermi energy? The answer is that there are states at the Fermi energy which are located near the edges of the smaple. An important point to note from Fig.1.4.2 is that at the minima the resistance is very nearly zero. This is particularly surprising since the voltage probes are typically spaced by hundreds of microns. The fact that the resistance is so close to zero shows that the electrons are able to travel such huge distances without losing their momentum. Clearly something rather special must be happening at microscopic level leading to this fantastic supperssion of momentum relaxation processes. In Section 1.6, while increasing the magnetic field in a finite-width conductor, the states carrying current in one direction get spatially separated from the states carrying current in the opposite direction. The result is a significant reduction in the spatial overlap between the forward and the backward propagating states which leads to a suppression of backscattering (and hence momentum relaxation). Indeed at high magnetic fields the forward and backward propagating states are spatially separated by the witdth of the conductor and thus have practically zero overlap in wide conductors. In Section 1.6 we assumed a parabolic confining potential $U(y) = m\omega_0^2 y^2/2$. This allowed us to obtain the eigenfunctions

of the Schrodinger equation

$$\left[E_s + \frac{(\mathrm{i}\hbar\nabla + e\boldsymbol{A})^2}{2m} + U(y)\right]\Psi(x,y) = E\Psi(x,y) \tag{4.1.1}$$

analytically. A parabolic potential often provides a good description of narrow quantum wires. But for wide conductors, the transverse confining potential usually looks more like that shown in Fig.4.1.1a. In general, analytical solutions are not available for arbitrary confining potentials. However, there is an approximate solution that we can use at high magnetic fields. It is quite accurate if the cyclotron radius is small enough that the confining potential can be assumed to be nearly constant on this scale.

4.1.1 Magneto-electric subbands at high magnetic fields

We know that if the confining potential were absent, then the solutions to the Schrodinger equation would given by

$$\begin{aligned} \Psi_{n,k}(x,y) &= \frac{1}{\sqrt{L}}\exp(ikx)u_n(q+q_k) \equiv |n,k\rangle \\ E(n,k) &= E_s + (n+\frac{1}{2})\hbar\omega_c, \quad n = 0,1,2,\cdots \end{aligned} \tag{4.1.2}$$

where

$$\begin{aligned} u_n(q) &= \exp\left(-q^2/2\right)H_n(q) \\ q &= \sqrt{m\omega_c/\hbar}y, \quad q_k = \sqrt{m\omega_c/\hbar}y_k \\ y_k &\equiv \frac{\hbar k}{eB}, \quad \omega_c \equiv \frac{|e|B}{m} \end{aligned} \tag{4.1.3}$$

$H_n(q)$ is the nth Hermite polynomial. Now let us use the lowest order perturbation theory to include the effect of the confining potential $U(y)$.

$$E(n,k) \approx E_s + (n+\frac{1}{2})\hbar\omega_c + \langle n,k|\, U(y)\, |n,k\rangle \tag{4.1.4}$$

Note that each state (n,k) is centered around a differnet localtion $y = y_k$ in the transverse direction and has a spatial extent of $\sim(\hbar/m\omega_c)^{1/2}$. Assuming that the potential $U(y)$ is nearly constant over the extent of each state, we can write

$$E(n,k) \approx E_s + (n+\frac{1}{2})\hbar\omega_c + U(y_k), y_k = \hbar k/eB \tag{4.1.5}$$

Figure 4.1.1b shows a sketch of the dispersion relation. It looks just like the confining potential $U(y)$, with the coordinate y mapped onto the wavenumber k by the relation $y_k = \hbar k/eB$. In the middle of the sample the states look just like the Landau levels of an unconfined 2D conductor spaced by $\hbar\omega_c$. Near the edges there are allowed states with a continuous distribution of energies. These are referred to as the edge states and they play a very important role in carrying the current at the resistance minimum.

We can calculate the velocity of an electron in the state (n,k) as follows:

$$v(n,k) = \frac{1}{\hbar}\frac{\partial E(n,k)}{\partial k} = \frac{1}{\hbar}\frac{\partial U(y_k)}{\partial k} = \frac{1}{\hbar}\frac{\partial U(y_k)}{\partial y_k}\frac{\partial y_k}{\partial k} = \frac{1}{eB}\frac{\partial U(y_k)}{\partial y_k} \tag{4.1.6}$$

Thus the states located at the two edges of the sample carry current in opposite directions since the slope of the confining potential is opposite at the two edges. The resistance of the sample is determined by the rate at which the electrons in states between μ_1 and μ_2 can relax their momentum. The statees carrying current in one direction are spatially separated from those carrying current in the opposite direction. To relax momentum an electron has to be scattered from the left of the sample to the right of the smaple. This is all but impossible since the overlap between the wavefunctions is exponentially small and there are no allowed states in the interior of the sample in this energy range ($\mu_1 > E > \mu_2$). As a result of this complete supperssion of backscattering, electron originating in the left contact enter the edge states carrying current to the right and empty into the right contact, while electrons in the right contact enter the edge states carrying current to the left and empty into the left contact.Consequently, the edge states carrying current to the right are completely in equilibrium with the left contact and have a quasi-Fermi energy equal to μ_L. They are unaffected by μ_R since no electron originating in the right contact ever makes it to these states. Similarly we can argue that the edge states carrying current to the left all originate from the right contact and have a quasi-Fermi energy equal to μ_R

$$\mu_1 = \mu_L, \quad \mu_2 = \mu_R \tag{4.1.7}$$

Clearly the longitudinal voltage drop V_L as measured by two voltage probes located anywhere on the same side of the sample is zero, while the transverse (or Hall) voltage V_H measured by two probes located anywhere on opposite sides of the sample is equal to the applied voltage:

$$V_L = 0, \quad eV_H = \mu_L - \mu_R \tag{4.1.8}$$

Note that this situation arises only when the electrochemical potentials lie between two bulk Landau levels. If the elecrochemical potentials lie on a bulk Landau level then there is a continuous distribution of allowed states from one edge to the other. Electrons can scatter from the left of the sample to the right of the sample through the allowed energy states in the interior of the sample. This backscattering gives rise to a maximum in the longitudinal resistance every time the Fermi energy lies on a bulk Landau level.

4.1.2 What is the current?

The number of edge states (which is equal to the number of filled Landau levels in the bulk) plays the role played by the number of modes in a ballistic conductor so that we can write

$$I = \frac{2e}{h} M(\mu_L - \mu_R) \tag{4.1.9}$$

We could derive this formally as follows (k_L and k_R are the wavenumbers corresponding to $E = \mu_L$ and $E = \mu_R$ respectively)

$$\begin{aligned} I &= 2e \sum_n \int_{\mu_R}^{\mu_L} \frac{1}{2\pi} v(n,k) \mathrm{d}k = 2e \sum_n \int_{k_R}^{k_L} \frac{1}{2\pi} \frac{1}{\hbar} \frac{\partial E(n,k)}{\partial k} \mathrm{d}k \\ &= \frac{2e}{h} \sum_n \int_{\mu_R}^{\mu_L} \mathrm{d}E = \frac{2e}{M} \left[\mu_L - \mu_R\right] \end{aligned} \tag{4.1.10}$$

Hence we can write down the longitudinal and Hall resistances:

$$R_L = \frac{V_L}{I} = 0, \quad R_H = \frac{V_H}{I} = \frac{h}{2e^2 M} \tag{4.1.11}$$

where M is the number of edge states at the Fermi energy, which is the number of bulk Landau levels below the Fermi energy. M takes on integer values that decrease as the magnetic field is increased. What is impressive is the striking accuracy of the quantization of R_H that is obtained at high magnetic fields.

4.2 Application of the Buttiker formalism

$$
\begin{array}{ccccccc}
G_{pq}: & q=1 & q=2 & q=3 & q=4 & q=5 & q=6 \\
p=1 & 0 & 0 & 0 & 0 & 0 & G_C \\
p=2 & G_C & 0 & 0 & 0 & 0 & 0 \\
p=3 & 0 & G_C & 0 & 0 & 0 & 0 \\
p=4 & 0 & 0 & G_C & 0 & 0 & 0 \\
p=5 & 0 & 0 & 0 & G_C & 0 & 0 \\
p=6 & 0 & 0 & 0 & 0 & G_C & 0
\end{array}
\tag{4.2.1}
$$

where $G_C = 2e^2M/h$. Setting $V_4 = 0$

$$
\begin{pmatrix} I_1 \\ I_2 \\ I_3 \\ I_5 \\ I_6 \end{pmatrix} = \begin{bmatrix} G_C & 0 & 0 & 0 & -G_C \\ -G_C & G_C & 0 & 0 & 0 \\ 0 & -G_C & G_C & 0 & 0 \\ 0 & 0 & 0 & G_C & 0 \\ 0 & 0 & 0 & -G_C & G_C \end{bmatrix} \begin{pmatrix} V_1 \\ V_2 \\ V_3 \\ V_5 \\ V_6 \end{pmatrix}
\tag{4.2.2}
$$

The currents at the voltage terminals are all zero$(I_2 = I_3 = I_5 = I_6 = 0)$

$$
V_2 = V_3 = V_1, \quad V_5 = V_6 = 0
\tag{4.2.3}
$$

Also the current is given by

$$
I_1 = G_C V_1
\tag{4.2.4}
$$

So that the longitudinal and Hall resistances are given by

$$
\begin{aligned}
R_L &= \frac{V_2 - V_3}{I_1} && = 0 \\
R_H &= \frac{V_2 - V_6}{I_1} && = \frac{h}{2e^2M}
\end{aligned}
\tag{4.2.5}
$$

4.2.1 Dose the current flow only at the edges?

We stated above that if $\mu_1 > \mu_2$ then the states below μ_2 are all filled and do not carry any net current. Any net current can be calculated from the filled states on the left between μ_1 and μ_2. However, this does not mean that currents everywhere in the sample. We could choose to do our bookkeeping in a differnet way so that the net current appears at a different spatial location. We have identified all the sates below μ_2 as our Fermi sea which does not carry any net current. Consequently the net current is carried by the electrons in the edge states on the side of the sample with energies lying in the range $\mu_1 > E > \mu_2$. But we could just as well identify all the states below μ_1 as our Fermi sea. The net current would then be carried by the holes occupying the edge staes on the other side of the sample! At low temperatures the current is carried by the states near the Fermi energy. States lying deep inside the Fermi sea have no effect on the conductance. But the local current density due to these states is not zero. These states give rise to circulating currents in the sample even at equilibrium. An applied electric field can induce a change in this circulating current flow pattern thus contributing to the confuctivity tensor defined by the relation $\delta \boldsymbol{J} = \sigma \delta E$. Thus electrons deep inside the Fermi sea can contribute to the conductivity, even though they do not contribute to th conductance.

4.2.2 Why should the Fermi energy ever lie between Landau levels?

The above discussion shows that the longitudinal resistance can be extremely small if the electrochemical potentials μ_1 and μ_2 were located between bulk Landau levels. This requires that the equilibrium Fermi energy E_f must be located between the Landau levels since at low bias

$\mu_1 \sim \mu_2 \sim E_f$. How is the location of E_f determined?
At low temperatures we can write

$$n_s = \int_{-\infty}^{E_f} N_s(E, B)\mathrm{d}E \tag{4.2.6}$$

where n_s is the electron density and N_s is the density of states. The electron density increases with Fermi energy as shown in Fig.4.1.4. The important point to note is that the electron density increases rapidly whenever the density of states is high. This is because a change in the electron density is related to the change in the Fermi energy by the relation

$$\delta n_s = N_s(E_f, B)\delta E_f \tag{4.2.7}$$

4.2.3 Fractional quantum Hall effect

We have seen that in the quantum Hall regime the Hall resistance takes on quantized values given by

$$\rho_{yx} = \frac{h}{2e^2 M}, \quad M = 1, 2, 3, \cdots \tag{4.2.8}$$

Actually at high fields the energy levels for the two spins split apart due to the Zeeman effect and quantized plateaus are obtained with

$$\rho_{yx} = \frac{h}{e^2 M} \tag{4.2.9}$$

When the magnetic field reaches a value such that the electron density $n_s = eB/h$, we will have all the electrons in a single Landau level with one spin. For a carrier density of $n_s = 2 \times 10^{11}/\mathrm{cm}^2$ this requires a field of about $8T$. What happens if we increase the field further?
From our earlier discussion we might expect that there will be no further plateaus with increasing magnetic field since the Fermi energy now lies in the middle of a Landau level. Experimentally, however in very pure samples on continues to observe plateaus in the Hall resistivity given by

$$\rho_{yx} = \frac{h}{e^2 p} \tag{4.2.10}$$

where p is a rational fraction like 1/3, 2/5, 3/7 etc. This phenomenon is called the fractional quantum Hall effect (FQHE) and cannot be explained using the one-electron picture presented so far. The explanation of the FQHE requires us to take into account the strong electron-electron interactions that occur when all the electrons are confined to the lowest Landau level. The basic idea is that the electrons can lower their mutual repulsion by forming a new type of collective state in which each electron is surrounded by a correlation hole (a region of reduced electron density) that keeps other electrons away. The resulting quasiparticles have fractional charge and give rise to the observed fractional quantum Hall effect.